



Lightweight single image super-resolution based on multi-path progressive feature fusion and attention mechanism

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Abstract

Single image super-resolution (SISR) is a fundamental image processing task, which aims to generate a high-resolution (HR) image from its low-resolution (LR) counterpart. Deep convolution neural networks (CNNs) have significantly improved the performance of SISR, and dominated the current research on SISR techniques. However, the performance improvement depends heavily on the size of the networks. In general, the deeper the networks, the better the performance, which limits their use to devices with limited computing and memory resources. The challenge is to find the optimal balance between network model complexity and SISR performance. In this paper, we propose a new lightweight SISR algorithm based on CNNs, which uses multi-path progressive feature fusion and attention mechanism. Our main contributions are as follows: (1) We propose a multi-path progressive feature fusion block (PFF), which can use the feature from the previous path to gradually guide the feature learning of the next path step by step in multiple paths. (2) We propose a multi-path feature attention mechanism (FAM), which can adaptively weigh the multi-path feature channels to be concatenated, improve the utilization of feature information and feature representation capability. The experimental results show that whether it is an objective measurement or a subjective measurement, our method is better than other similar state of the art methods, and has a better model complexity and performance balance.

Keywords Image super-resolution · Convolutional neural network · Feature fusion · Attention mechanism

1 Introduction

Single image super-resolution (SISR) [8] is a fundamental image processing task, which aims to generate a high-resolution (HR) image from its low-resolution (LR) counterpart, and has a wide range of applications in remote sensing [41], video surveillance [50], medical imaging [37] and so on.

However, SISR is difficult, because it is an inherently ill-posed inverse problem. One LR image can correspond to multiple HR images [36] and the prior knowledge is needed to regularize the solution space.

SISR has a long history of research [34]. In recent years, deep convolutional neural networks (CNNs) have dominated the research of SISR methods [6, 7, 20–22, 39, 56], which learn mapping functions between LR and HR images directly end-to-end, and significantly improve the performance of SISR. However, CNNs-based SISR methods are heavily dependent on the size of the networks, i.e., the number of parameters, depth (number of layers) and width (number of channels) in the networks. To improve the performance of SISR, larger networks are often required. For example, EDSR [27] has 65 convolutional layers and about 43 M parameters; RCAN [55] has over 800 convolutional layers and about 16 M parameters. Although methods such as EDSR and RCAN have good performance, they require high computing and storage cost and are difficult to apply to resource-constrained devices (e.g., mobile phones). Designing lightweight networks (with low computing and storage

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requirements) is a hot topic of current SISR research. The challenge is to establish a better balance between the model complexity and the performance of the SISR methods.

Multi-path SISR network architectures are very popular. Zhao et al. [57] propose a very lightweight SISR network, called the pixel attention network (PAN). The basic building block of PAN (i.e., SC-PA block) contains two branches: one branch contains pixel attention (PA), which is similar to channel attention [14] and spatial attention [47] for higher-level feature manipulation. The other branch uses a standard convolutional layer to maintain the original information. Then, the output feature channels of the two branches are concatenated along the channel dimension. The research of Wang et al. [44] shows that the features of the previous layers (i.e., auxiliary features) can help the learning of the features of the current layer. The basic building block of their proposed A²F model is the A²F block, which also consists of two main branches: one branch uses the features of all previous blocks to compute the channel attention, the other branch computes the feature using the WDSR A block [49], and then the features of both branches and the original feature are weighted and concatenated together. The research of Zhao et al. [58] shows that the weighted channel concatenation is more in line with the behavior of human brain neurons and can improve the expressive ability of the features. Inspired mainly by the works of Wang et al. [44], Zhao et al. [57, 58], we propose a lightweight SISR network based on multi-path progressive feature fusion and attention mechanism (MPFFA). The core building block of MPFFA is the multi-path feature extraction block (MPFE), as shown in Fig. 2d. MPFE includes two sub-blocks: multi-path progressive feature fusion (PFF) and multi-path feature attention mechanism (FAM). Our MPFE is somewhat more similar to A²F, but, unlike A²F, the PFF uses the feature from the previous branch in the current block to gradually guide the feature learning of the next branch to predict the importance of the features pixel by pixel, without using the features from the previous blocks, as in A²F. In addition, when FAM concatenates the features of each branch along the channel dimension, the weights of the feature channels in each branch are computed by the channel attention mechanism, instead of using a scalar weight, which can more accurately weight the feature information. Experimental results show that our MPFFA is very effective and outperforms the other state-of-the-art lightweight SISR methods, taking the model complexity and performance balance to a new level (please refer to Fig. 1).

In summary, our contributions include: (1) We propose a novel lightweight SISR network (MPFFA) based on multi-path progressive feature fusion and attention mechanism, which outperforms the current state of the art lightweight SISR methods with better model complexity and performance balance; (2) We propose a multi-path progressive

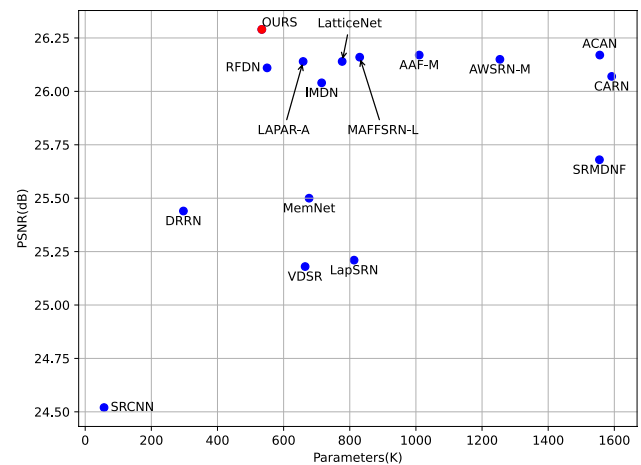


Fig. 1 Average PSNRs and numbers of parameters for 4× SR models for each state of the art SISR method on the Urban100 dataset. The horizontal and vertical axes represent the number of parameters (K) and PSNR (dB), respectively. The red dot represents our MPFFA, and the blue dots are the other methods

feature fusion block (PFF), which can use the previous features to guide the learning of subsequent multi-path features gradually; (3) We propose a multi-path feature attention mechanism (FAM), which can better weight and concatenate the multi-path features, and thus enhance the expression ability of features.

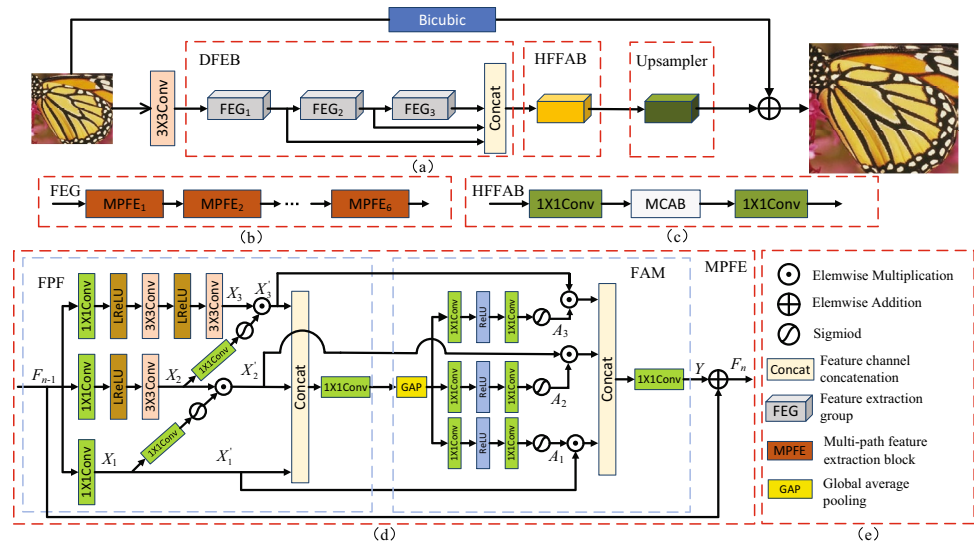
The remainder of this paper is organized as follows. In Sect. 2 the related works are presented. Section 3 describes the proposed methods. Section 4 reports experimental results and Sect. 5 concludes the paper.

2 Related works

2.1 Lightweight SISR networks

Dong et al. [6] propose the seminal CNNs-based SISR method, called SRCNN. Recent years have witnessed a surge of CNNs-based SISR methods [6, 7, 20–22, 39, 56]. Currently, CNNs have dominated the research of SISR methods. The CNNs-based SISR methods usually use L1 or L2 loss, and one of its problems is that the generated SR image is too smooth and lacks high-frequency details [25]. Generative Adversarial Networks (GANs) [10] are also introduced into SISR [25], which use perceptual and adversarial losses. GANs-based SISR methods [25, 59] can recover clearer high-frequency details, however, the generated SR images have low fidelity, severe distortion, and are also difficult to train. One of the other major problems of CNNs-based SISR methods is the heavy dependence on the size of the networks. The larger the network size, the stronger the expressive power and the

Fig. 2 **a** The network architecture of our proposed MPFFA. **b** Feature extraction group (FEG). **c** Hierarchical feature fusion attention block (HFFAB). **d** Multi-path feature extraction block (MPFE). **e** Sign description



better the performance, which also limits their practicality, especially for devices with limited computational and memory resources. Lightweight SISR methods based on CNNs, have attracted widespread attention. The challenge is to find the best balance between model complexity and performance. In lightweight SISR network structure design, the techniques of recursion are very popular. The DRCN proposed by Kim et al. [21] introduces recursion into the CNNs-based SISR methods for the first time. The parameters are shared among the convolutional layers, which can significantly reduce the model parameters. Tai et al. [39] propose a very deep persistent memory network (MemNet), which uses residual blocks as recursive units. The ACAN proposed by Zhou et al. [60] also uses residual blocks as recursive units. Although the recursive structure can reduce parameters, it cannot reduce the computational cost. 1×1 convolution, group convolution [3], depthwise separable convolution [13], and self-calibrating convolution [29], etc. are also widely used in lightweight SISR network architectures. Zhao et al. [57] propose a very lightweight SISR model, called PAN, which uses self-calibration convolutions as the basic building blocks, and also uses 1×1 convolution to design a lightweight pixel attention block. The basic building block of the IDN proposed by Hui et al. [18] is the information distillation block, which uses 1×1 convolution to compress and distill feature information. Hui et al. [19] also propose the information multi-distillation network (IMDN), which uses the information multi-distillation block (IMDB) as the basic building blocks. IMDB improves DBlock, using multi-level channel splitting, and processive refinement. The RFDN proposed by Liu et al. [28] further improves the channel splitting and feature distillation of IMDN to make it more lightweight and more effective. In this paper, our model adopts a novel multi-path progressive feature

fusion attention mechanism, which significantly improves the complexity and performance balance of current SISR models.

2.2 Multi-pathing learning

Multi-path learning is an important paradigm of the CNNs-based SISR network architectures. Residual connections [20] and dense connections [15] are probably the most common multi-path learning architectures. They can enhance the propagation of feature information, but also suffer from serious feature information redundancy. How to improve multi-path learning and increase the utilization and expressive ability of feature information is an important concern for the SISR research community [11, 17, 19, 28, 44, 57]. Han et al. [11] propose a dual-state recurrent network (DSRN), which extracts features simultaneously in LR and HR spaces through two branches and continuously interacts with each other to enhance the expression ability of features. Hui et al. [19] propose an information multi-distillation block (IMDB) which progressively extracts and refines the hierarchical features by channel splitting, with one branch maintaining the original feature information and the other branch performing convolutional operations. Liu et al. [28] propose a RFDB block, using the feature distillation connection (FDC), instead of the channel splitting in the IMDB block, which is more lightweight and more flexible. Huang et al. [17] propose a dual-path attention network (DPAN), the basic building block of which is the dual-path block (DPB), in which one branch performs residual learning and the other branch performs channel concatenation to better fuse the complementary feature information. The attentive auxiliary feature block (A²F) proposed by Wang et al. [44] contains two main branches: one branch projects the features of the preceding layers; the other branch uses the channel attention

mechanism, which learns the importance of each channel. Zhao et al. [57] propose a pixel attention network (PAN), the basic building block of which is the self-calibrated block with pixel attention (SC-PA). The SC-PA block consists of two branches: one branch uses pixel-attention (PA) to perform high-level feature manipulation, and the other branch uses standard convolutional layers to maintain the original information. Our model also uses a multi-path (branch) structure, with the main innovation being that the feature from one branch is used to progressively guide and calibrate the feature learning in the next branch.

2.3 Attention mechanism

Attention mechanism [4] usually refers to the human visual system, which can automatically focus on the salient areas. The attention mechanism in CNN network architectures is similar, which means that it can focus on specific parts of the input, which can improve the utilization and expressive ability of features. Hu et al. [14] propose a squeeze-and-excitation network (SENet), where the basic building block is squeeze-and-excitation (SE) blocks. The SE block uses global average pooling (GAP) to squeeze each input channel into channel descriptors (constants), which are then fed into two fully-connected layers to generate the weights for each input channel to explicitly model the interdependencies between channels. The residual channel attention network (RCAN) proposed by Zhang et al. [55] introduces the SENet mechanism [14] of Hu et al. into SISR. Dai et al. [5] further propose a second-order channel attention block, which adaptively rescales each feature channel by using second-order feature statistics, instead of GAP, and can better learn the correlations between feature channels. Woo et al. [47] introduce both channel attention and spatial attention mechanisms. It is possible to focus not only on the feature channels, but also on the spatial positions in the channels. Zhang et al. [52] propose a multi-context attentive block (MCAB), which can also focus on the dependencies between channel and spatial dimensions at the same time. In this paper, we propose a multi-path attention mechanism that can adaptively weight the multi-path features. We also use MCAB [52] to fuse the hierarchical features from the model.

3 Proposed method

The network architecture of our MPFFA is shown in Fig. 2a. It mainly includes four parts: Shallow feature extraction block (SFEB), Deep feature extraction block (DFEB), Hierarchical feature fusion attention block (HFFAB) and Upsampler. SFEB only has a 3×3 convolutional layer to extract shallow features. Upsampler

uses the upsampling block of PAN [57]. HFFAB mainly uses the multi-context attentive block (MCAB) in TSAN [52], as shown in Fig. 2c. To reduce the computational effort, the number of feature channels input to MCAB is compressed four times by 1×1 convolution, while the number of feature channels output from MCAB is expanded four times by 1×1 convolution. Our work is primarily focused on DFEB, which includes three feature extraction groups (FEG). Each FEG contains six multi-path feature extraction blocks (MPFE), as shown in Fig. 2b. Each MPFE consists of a multi-path progressive feature fusion block (PFF) and a multi-path feature attention mechanism (FAM), as shown in Fig. 2d. MPFFA also contains a global residual connection, where the LR image is enlarged to the size of the target SR image by bicubic interpolation.

Assuming that the input LR image is I_{LR} , it is first fed into the SFEB block to extract the shallow feature:

$$F_0 = H_{SFEB}(I_{LR}) \quad (1)$$

where $H_{SFEB}(\cdot)$ is the SFEB function, and F_0 is its output feature. F_0 is then fed into the DFEB to further extract the deep features. DFEB contains three FEGs, and each FEG contains six MPFEs. Therefore, F_0 passes through eighteen MPFEs, and the output of each MPFE can be expressed as:

$$F_n = H_n(F_{n-1}), (n = 1, 2, \dots, 18) \quad (2)$$

where $H_n(\cdot)$ is the n -th MPFE function, and F_n is its output feature. Considering that hierarchical features can provide complementary feature information [60], the hierarchical features F_6 , F_{12} and F_{18} output by the three FEGs are concatenated to improve feature utilization and expressiveness:

$$F_{cat} = [F_6, F_{12}, F_{18}] \quad (3)$$

where $[\cdot]$ is the feature channel concatenation operation, and F_{cat} is the concatenated feature, which is also the output feature of the DFEB block, and then is fed into HFFAB:

$$F_{hff} = H_{HFFAB}(F_{cat}) \quad (4)$$

where $H_{HFFAB}(\cdot)$ is the HFFAB function, and F_{hff} is its output feature, which is then fed into Upsampler block:

$$I_{SR} = H_{Up}(F_{hff}) + H_{Bic}(I_{LR}) \quad (5)$$

where $H_{Up}(\cdot)$ is the Upsampler function, and $H_{Bic}(\cdot)$ is the bicubic interpolation function, which is added to the output of the residual $H_{Up}(\cdot)$ to obtain the final output of the target SR image I_{SR} . In the current SISR methods, most upsampling blocks use sub-pixel convolution [27] or transposed convolution [9], here we use the upsampling block of PAN [57].

3.1 Multi-path progressive feature fusion block (PFF)

The multi-path progressive feature fusion block (PFF), mainly inspired by the works of Wang et al. [44] and Zhao et al. [57], performs multi-path (we use three paths) progressive feature fusion, which uses the feature of the previous branch to guide and calibrate the feature learning of the next branch. As shown on the left side of Fig. 2d, the bottom branch contains only a 1×1 convolutional layer. The middle branch contains a 1×1 convolutional layer and a 3×3 convolutional layer, with an LReLU [12] nonlinear activation function between the two convolutional layers. The feature of the bottom branch guides and calibrates the feature of the middle branch by the 1×1 convolution and Sigmoid function. The top branch contains a 1×1 convolutional layer and two 3×3 convolutional layers, and there is an LReLU nonlinear activation function between every two convolutional layers. The feature of the middle branch guides and calibrates the feature of the top branch by the 1×1 convolution and the Sigmoid function again. Considering that F_{n-1} and F_n are the input and output of the n -th MPFE, respectively, as shown in Eq. (2). The output features of the three branches before calibration are as follows (ignoring the LReLU nonlinear activation functions):

$$X_1 = \text{Conv}_{1 \times 1}(F_{n-1}) \quad (6)$$

$$X_2 = \text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F_{n-1})) \quad (7)$$

$$X_3 = \text{Conv}_{3 \times 3}(\text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F_{n-1}))) \quad (8)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, and $\text{Conv}_{3 \times 3}(\cdot)$ is the 3×3 convolution function. The output features of the three branches after calibration can be described as:

$$X'_1 = X_1 \quad (9)$$

$$X'_2 = X_2 \odot (\sigma(\text{Conv}_{1 \times 1}(X_1))) \quad (10)$$

$$X'_3 = X_3 \odot (\sigma(\text{Conv}_{1 \times 1}(X_2))) \quad (11)$$

where $\sigma(\cdot)$ is the Sigmoid function, and the $\odot$ is the element-wise multiplication. Then, the calibrated output feature channels of the three branches are concatenated, and the channels are compressed and fused with a 1×1 convolutional layer:

$$X' = \text{Conv}_{1 \times 1}([X'_1, X'_2, X'_3]) \quad (12)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, and $[\cdot]$ is the feature channel concatenation operation.

3.2 Multi-path feature attention mechanism (FAM)

The multi-path feature attention mechanism (FAM) is mainly inspired by the works of Wang et al. [44] and Zhao et al. [57], which indicate that the weighted concatenation of feature channels can improve the expressive ability of features. Unlike their works, instead of using scalar weights, the FAM uses the channel attention mechanism to optimally weight the channels of the features from each branch of the PFF and then concatenate them. The output feature X' of the n -th PFF block, through global average pooling (GAP), generates an initial weight vector. Then, the weight vector is processed by three branches. The feature channels of each branch are compressed and expanded by two 1×1 convolutions (with a ReLU [1] nonlinear activation function between the two 1×1 convolutions). Next, using the Sigmoid function generates the weight vector of the feature channels of each branch.

Assuming the concatenated feature output by the n -th PFF block $X' = [x_1, \dots, x_c, \dots, x_C] \in \mathbb{R}^{H \times W \times C}$ (H , W and C are the height, width and number of channels of the feature, respectively), the GAP can be expressed as:

$$z_c = H_{GAP}(x_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j) \quad (13)$$

where $H_{GAP}(\cdot)$ is the GAP function, and $Z = [z_1, \dots, z_c, \dots, z_C] \in \mathbb{R}^C$. The weight of each channel of the three branches can be expressed as (ignoring the ReLU nonlinear activation function):

$$A_k = \sigma(\text{Conv}_{1 \times 1}(\text{Conv}_{1 \times 1}(Z))), (k = 1, 2, 3) \quad (14)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, and $\sigma(\cdot)$ is the Sigmoid function. The residual output Y of the n -th FAM can be expressed as:

$$Y = \text{Conv}_{1 \times 1}([A_1 \odot X'_1, A_2 \odot X'_2, A_3 \odot X'_3]) \quad (15)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, $X'_k (k = 1, 2, 3)$ is the output feature of the k -th branch of the n -th PFF block, $[\cdot]$ is the feature channel concatenation operation, and $\odot$ is the element-wise multiplication. Since the residual connection is used, the output feature F_n of the n -th FAM needs to be added to the input F_{n-1} of the n -th PFF block:

$$F_n = Y + F_{n-1} \quad (16)$$

3.3 Hierarchical feature fusion attention block (HFFAB)

The multi-context attentive block (MCAB) [52] can focus on information-rich contextual features. The hierarchical

feature fusion attention block (HFFAB) only adds the 1×1 convolutional layers before and after the MCAB, as shown in Fig. 2(c). The preceding 1×1 convolutional layer compresses the feature channels to reduce the computational effort; the following 1×1 convolutional layer then expands the feature channels. The operation of HFFAB block, see Eq. (4), can be further expressed as:

$$F_{hff} = \text{Conv}_{1 \times 1}(H_{MCAB}(\text{Conv}_{1 \times 1}(F_{cat}))) \quad (17)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, $H_{MCAB}(\cdot)$ is the MCAB block function, and F_{cat} is the output feature of the DFEB block.

4 Experiments

4.1 Settings

We used the popular DIV2K [42] dataset as the training and validation data. The first 800 images (001–800) were used for training, and 10 images (821–830) were used for validation, labeled as DIV2K_val10. The original HR training images were bicubically downsampled to obtain the paired LR training images. Similar to other methods, the training images were augmented by randomly rotating 90° , 180° , 270° and flipping horizontally. Our test datasets were five standard benchmark ones, Set5 [3], Set14 [51], B100 [31], Urban100 [16], and Manga109 [32]. As with the previous methods, the SR results were evaluated by PSNR and SSIM [46] metrics on the luminance (Y) channel of the transformed YCbCr space [56]. We first trained the $2\times$ model and then train the $3\times$ and $4\times$ models based on the pre-trained $2\times$ model.

In the model training, 32 patches with the size of 48×48 were randomly selected for each batch. The Adam [23] optimizer was used, and set $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$. The model was trained for 1200 epochs with an initial learning rate set to 2×10^{-4} and halved for every 200 epochs. The L1 loss function was used. The model was implemented by using PyTorch [35] with a NVIDIA RTX 2080Ti GPU.

4.2 Ablation study

In order to evaluate the effects of the multi-path progressive feature fusion block (PFF), the multi-path feature attention mechanism (FAM) and the hierarchical feature fusion attention block (HFFAB), we conducted four comparative experiments on DIV2K_val10 validation dataset. The first experiment did not include FAM and HFFAB blocks. The PFF block also contained only three branches, and did not include the calibration of the previous features to the subsequent features, that is, it did not include the 1×1 convolution and Sigmoid function between the two branches. This experiment was called MPFFA_0. In the second experiment, the calibration of the previous features to the subsequent features was added to MPFFA_0. That is, the PFF block was completely included. However, the FAM and HFFAB blocks were still not included. This experiment was called MPFFA_1. The third experiment was to add the HFFAB block to MPFFA_1, which was called MPFFA_2. The fourth experiment was to add the FAM block to MPFFA_2, which was our complete MPFFA network model. The PSNRs of the four models on the DIV2K_val10 validation dataset with $3\times$ SR were shown in Table 1. It could be seen that the PSNR of MPFFA_0 was only 29.49 dB and the PSNR of MPFFA_1 was 0.11 dB higher than MPFFA_0. It seemed to indicate that the multi-path progressive feature fusion, which used the previous features assisting the learning of the subsequent features, was effective. Compared with MPFFA_1, the PSNR of MPFFA_2 also increased by 0.03 dB. Compared with MPFFA_2, the PSNR of the final model MPFFA increased by 0.06 dB, which indicated that the FAM block using the weighted concatenation of multi-path feature channels was also effective.

4.3 Experiments on the number of MPFEs

To ensure that our model was effective enough and light-weight enough, three feature extraction groups (FEGs) were used, and each FEG contained six multi-path feature extraction blocks (MPFE). The selection of the number of MPFEs needed to consider the impacts on the number of parameters, the computational effort, and the performance. We conducted three experiments on the DIV2K_val10 validation

Table 1 Average PSNRs, number of parameters and computational cost of $3\times$ SR on validation dataset DIV2K_val10 for different variants of network architecture

Model	PFF's calibration	PFF's three branches	HFFAB	FAM	PSNR (dB)	Parameters (K)	FLOPs (G)
MPFFA_0	✓	×	×	×	29.49	443	59.4
MPFFA_1	✓	✓	×	×	29.60	465	61.6
MPFFA_2	✓	✓	✓	×	29.63	480	63.0
MPFFA	✓	✓	✓	✓	29.69	523	66.2

dataset. In each experiment, the number of MPFEs in each FEG was set to 5, 6 and 7, respectively. The PSNR results for 3× SR were shown in Table 2. It could be seen that increasing the number of MPFEs could improve the performance of the model, but the number of parameters also increased. It was assumed that the number of MPFEs was represented by n . Compared to $n = 5$, when $n = 6$, the number of parameters increased by 78 K and the PSNR increased by 0.14 dB. Compared to $n = 7$, when $n = 6$, the number of parameters was 78 K less and the PSNR reduced by only 0.05 dB. $n = 6$ was probably a reasonable compromise.

4.4 Experiments on FAM

4.4.1 Experiments on the number of branches

To gain further insight into the effect of multi-paths in the FAM, we experimented with the case of the FAM with only one branch on the DIV2K_val10 validation dataset. The PSNR of the 3× SR was 29.56 dB, which was 0.13 dB lower than the PSNR when the FAM contained three branches. This result was interesting because when the FAM contained only one branch, the PSNR was 0.07 dB lower than the that of MPFFA_2 (without FAM block).

4.4.2 Experiments with shared parameters for each branch

Parameter sharing can reduce the number of parameters. If the parameters of each branch were shared in the FAM, the PSNR of 3× SR was 29.65 dB on the validation dataset DIV2K_val10, which was 0.04 dB lower than that of the case where the parameters of each branch were not shared.

4.5 Comparisons with state-of-the-arts

In order to verify the effectiveness of our MPFFA method, we conducted objective quantitative comparisons and subjective visual effect comparisons with the other state-of-the-art methods, which included SRCNN[6], FSRCNN [7], VDSR [20], DRCN [21], LapSRN [24], DRRN [40], MemNet [39], SRM-DNF [53], CARN [2], IMDN [19], LAPAR-A [26], RFDN [28], A²F-M [44], ACAN [60], LatticeNet [30], AWSRN-M [43], PAN [57], MAFFSRN-L [33], EMASRN [61] and ShuffleMixer [38]. On five standard test datasets, the average

PSNR/SSIM results for 2×, 3× and 4× SR were shown in Table 3, the visual comparisons were shown in Fig. 3. Except for LatticeNet and PAN, the PSNR/SSIM results of the other methods were obtained from the the original papers. The results of LatticeNet were derived from running the trained model provided by the author [48]. Since the original PAN uses both DIV2K and Flickr2K [45] training datasets, for the fair comparisons, we used the source code provided by the authors and re-trained it on the DIV2K training dataset. Therefore, the test results were slightly different from those in the original paper. Table 3 also included the number of parameters and GFLOPs [54] for each state of the art method. GFLOPs refers to the number of multiplication-addition operations.

Quantitative comparisons: As seen in Table 3, our PSNR/SSIM results were significantly better than the other methods. For example, the cutting-edge LAPAR-A and RFDN were slightly larger than our MPFFA model, but our results were almost always better than theirs. For 2×, 3× and 4× SR on each dataset, our maximum increases in PSNR were 0.31 dB, 0.34 dB, and 0.29 dB for LAPAR-A, and 0.22 dB, 0.18 dB, and 0.18 dB for RFDN, respectively. The number of parameters of A²F-M was about twice ours, but the PSNR results of 2×, 3× and 4× SR on the Manga109 dataset were 0.11 dB, 0.19 dB and 0.14 dB lower than ours, respectively. For 4× SR, the number of parameters of ACAN was about three times higher than ours, but the PSNR on the Manga109 dataset was 0.18 dB lower than ours, while the computational effort of ACAN was about 26 times higher than ours. We also gave the 4× SR inference times of several representative models in Table 4. It was interesting to note that our model had the advantages of PSNR/SSIM performance and computational cost, however, there seemed to be no significant advantage in the inference time, which would be an issue for our further study.

Visual comparisons: Figure 3 showed the 4× SR images of each method. Our results were better than those of other methods, and were significantly visible. Taking the image img067 in Urban100 dataset as an example, the stripes recovered by Bicubic, EMASRN, ShuffleMixer, LatticeNet and A²F-M etc. had severe blurring or distortion. Our results were very close to the original HR image. For the image img089 in the Urban100 dataset, the stripes recovered by other methods were either excessively blurred, or the direction of the stripes was wrong, while we recovered the stripes in the right direction, and also much clearer. The results for the image barbara in the Set14 dataset were also similar.

Table 2 PSNRs and numbers of parameters for 3× SR experiments on DIV2K_val10 for different number of the MPFEs in each FEG

Number of MPFE	PSNR(dB)	Parameters (K)
5	29.55	445
6	29.69	523
7	29.74	601

5 Conclusions

In this paper, we propose a novel lightweight SISR algorithm based on multi-path progressive feature fusion and attention mechanism. Our main work is to propose a multi-path

Table 3 Average PSNR/SSIM results of 2×, 3× and 4× SR for various state-of-the-art SISR methods, with the best and second-best results marked in bold and italics, respectively

Scale	Model	PRMs	FLOPs	Set5	Set14	B100	Urban100	Manga109
×2	SRCNN	57	52.7	36.66/0.9524	32.42/0.9063	31.36/0.8879	29.50/0.8946	35.74/0.9661
	FSRCNN	12	6.0	37.00/0.9558	32.63/0.9088	31.53/0.8920	29.88/0.9020	36.67/0.9694
	VDSR	665	612.6	37.53/0.9587	33.03/0.9124	31.90/0.8960	30.76/0.9140	37.22/0.9729
	DRCN	1774	17974.3	37.63/0.9588	33.04/0.9118	31.85/0.8942	30.75/0.9133	37.63/0.9723
	LapSRN	813	29.9	37.52/0.9590	33.08/0.9130	31.80/0.8950	30.41/0.9100	37.27/0.9740
	DRRN	297	6796.9	37.74/0.9591	33.23/0.9136	32.05/0.8973	31.23/0.9188	37.92/0.9760
	MemNet	677	2662.4	37.78/0.9597	33.28/0.9142	32.08/0.8978	31.31/0.9195	37.72/0.9740
	SRMDNF	1513	347.7	37.79/0.9600	33.32/0.9150	32.05/0.8980	31.33/0.9200	38.07/0.9761
	CARN	1592	222.8	37.76/0.9590	33.52/0.9166	32.09/0.8978	31.92/0.9256	38.36/0.9765
	IMDN	694	158.8	38.00/0.9605	33.63/0.9177	32.19/0.8996	32.17/0.9283	38.88/ 0.9774
	LAPAR-A	548	171.0	38.01/0.9605	33.62/0.9183	32.19/0.8999	32.10/0.9283	38.67/0.9772
	RFDN	534	123.0	38.05/0.9606	33.68/0.9184	32.16/0.8994	32.12/0.9278	38.88/0.9773
	A ² F-M	999	224.2	38.04/0.9607	33.67/0.9184	32.18/0.8996	32.27/0.9294	38.87/ 0.9774
	ACAN	800	2108.0	38.10/0.9608	33.60/0.9177	32.21/0.9001	32.29/0.9297	38.81/0.9773
	LatticeNet	756	169.5	38.06/0.9607	33.70/0.9187	32.20/0.8999	32.25/0.9288	38.94/ 0.9774
	AWSRN-M	1063	244.1	38.04/0.9605	33.66/0.9181	32.21/0.9000	32.23/0.9294	38.66/0.9772
	PAN	261	70.5	37.99/0.9603	33.53/0.9174	32.14/0.8992	31.93/0.9263	38.37/0.9766
	MAFFSRN-L	790	154.4	38.07/0.9607	33.59/0.9177	32.23/0.9005	32.38/0.9308	—/—
	ShuffleMixer	394	91	38.01/0.9606	33.63/0.9180	32.17/0.8995	31.89/0.9257	38.83/ 0.9774
	MPFFA(ours)	523	129.4	38.09/0.9607	33.70/0.9190	32.21/0.9000	32.34/0.9296	38.98/0.9767
×3	SRCNN	57	52.7	32.75/0.9090	29.28/0.8209	28.41/0.7863	26.24/0.7989	30.59/0.9107
	FSRCNN	12	5.0	33.16/0.9140	29.43/0.8242	28.53/0.7910	26.43/0.8080	30.98/0.9212
	VDSR	665	612.6	33.66/0.9213	29.77/0.8314	28.82/0.7976	27.14/0.8279	32.01/0.9310
	DRCN	1774	17974.3	33.82/0.9226	29.76/0.8311	28.80/0.7963	27.15/0.8276	32.31/0.9328
	DRRN	297	67969.0	34.03/0.9244	29.96/0.8349	28.95/0.8004	27.53/0.8378	32.74/0.9390
	MemNet	677	2662.4	34.09/0.9248	30.00/0.8350	28.96/0.8001	27.56/0.8376	32.51/0.9369
	SRMDNF	1530	156.3	34.12/0.9250	30.04/0.8370	28.97/0.8030	27.57/0.8400	33.00/0.9403
	CARN	1592	118.8	34.29/0.9255	30.29/0.8407	29.06/0.8034	27.38/0.8404	33.50/0.9440
	IMDN	703	71.5	34.36/0.9270	30.32/0.8417	29.09/0.8046	28.17/0.8519	33.61/0.9445
	LAPAR-A	594	114.0	34.36/0.9267	30.34/0.8421	29.11/0.8054	28.15/0.8523	33.51/0.9441
	RFDN	541	55.4	34.41/0.9273	30.34/0.8420	29.09/0.8050	28.21/0.8525	33.67/0.9449
	A ² F-M	1003	100.0	34.50/0.9278	30.39/0.8427	29.11/0.8054	28.28/0.8546	33.66/0.9453
	ACAN	1115	1051.7	34.46/0.9277	30.39/ 0.8435	29.11/0.8055	28.28/0.8550	33.61/0.9447
	LatticeNet	765	76.3	34.40/0.9272	30.32/0.8416	29.10/0.8049	28.19/0.8513	33.63/0.9442
	AWSRN-M	1143	116.6	32.42/0.9275	30.32/0.8419	29.13/0.8059	28.26/0.8545	33.64/0.9450
	PAN	261	39.0	34.30/0.9266	30.30/0.8416	29.06/0.8042	28.02/0.8493	33.34/0.9435
	MAFFSRN-L	807	68.5	34.45/0.9277	30.40/0.8432	29.13/0.8061	28.26/ 0.8552	—/—
	EMASRN	427	854.5	34.36/0.9264	30.30/0.8411	29.05/0.8035	28.04/0.8493	33.43/0.9433
	ShuffleMixer	415	43	34.40/0.9272	30.37/0.8423	29.12/0.8051	28.08/0.8498	33.69/0.9448
	MPFFA(ours)	523	66.2	34.51/0.9283	30.37/0.8424	29.13/0.8058	28.35/0.8550	33.85/0.9461

Table 3 (continued)

Scale	Model	PRMs	FLOPs	Set5	Set14	B100	Urban100	Manga109
×4	SRCNN	57	52.7	30.48/0.8628	27.49/0.7503	26.90/0.7101	24.52/0.7221	27.66/0.8505
	FSRCNN	12	4.6	30.71/0.8657	27.59/0.7535	26.98/0.7150	24.62/0.7280	27.90/0.8517
	VDSR	665	612.6	31.35/0.8838	28.01/0.7674	27.29/0.7251	25.18/0.7524	28.83/0.8809
	DRCN	1774	17974.3	31.53/0.8854	28.02/0.7670	27.23/0.7233	25.14/0.7510	28.98/0.8816
	LapSRN	813	149.4	31.54/0.8850	28.19/0.7720	27.32/0.7280	25.21/0.7560	29.09/0.8845
	DRRN	297	6796.9	31.68/0.8888	28.21/0.7720	27.38/0.7284	25.44/0.7638	29.46/0.8960
	MemNet	677	2662.4	31.74/0.8893	28.26/0.7723	27.40/0.7281	25.50/0.7630	29.42/0.8942
	SRMDNF	1555	89.3	31.96/0.8930	28.35/0.7770	27.49/0.7340	25.68/0.7730	30.09/0.9024
	CARN	1592	90.9	32.13/0.8937	28.60/0.7806	27.58/0.7349	26.07/0.7837	30.47/0.9084
	IMDN	715	40.9	32.21/0.8948	28.58/0.7811	27.56/0.7353	26.04/0.7838	30.45/0.9075
	LAPAR-A	659	94.0	32.15/0.8944	28.61/0.7818	27.61/0.7366	26.14/0.7871	30.42/0.9074
	RFDN	550	31.6	32.24/0.8952	28.61/0.7819	27.57/0.7360	26.11/0.7858	30.58/0.9089
	A ² F-M	1010	56.7	32.28/0.8955	28.62/0.7828	27.58/0.7364	26.17/0.7892	30.57/0.9100
	ACAN	1556	616.5	32.24/0.8955	28.62/0.7824	27.59/0.7366	26.17/0.7891	30.53/0.9086
	LatticeNet	777	43.6	32.18/0.8943	28.61/0.7812	27.57/0.7355	26.14/0.7844	30.54/0.9075
	AWSRN-M	1254	72.0	32.21/0.8954	28.65/0.7832	27.60/0.7368	26.15/0.7884	30.56/0.9093
	PAN	272	28.2	32.06/0.8939	28.56/0.7813	27.55/0.7352	26.01/0.7830	30.33/0.9069
	MAFFSRN-L	830	38.6	32.20/0.8953	28.62/0.7822	27.59/0.7370	26.16/0.7887	-/-
	EMASRN	546	1054.4	32.17/0.8948	28.57/0.7809	27.55/0.7351	26.01/0.7838	30.41/0.9076
	ShuffleMixer	411	28	32.21/0.8953	28.66/0.7827	27.61/0.7366	26.08/0.7835	30.65/0.9093
	MPFFA(ours)	534	43.0	32.36/0.8968	28.69/0.7835	27.61/0.7372	26.29/0.7902	30.71/0.9115

The number of parameters(K) and FLOPs(G) for each method were also included

Table 4 Inference times of 4× SR for different models (Low-resolution images were all super-resolved to 1280×720 size)

Model	Time (ms)
VDSR	131.78
DRRN	363.29
MemNet	45.70
CARN	26.44
LAPAR-A	33.34
RFDN	10.29
A ² F-M	17.86
ACAN	117.51
LatticeNet	23.17
AWSRN-M	17.78
EMASRN	525.57
ShuffleMixer	28.56
MPFFA(ours)	48.46

progressive feature fusion (PFF) and a multi-path feature attention mechanism (FAM), where PFF and FAM blocks are used in pairs. PFF can use the previous features to progressively guide and assist the subsequent feature learning, while FAM can weightedly concatenate the multi-path feature channels to improve the feature utilization and expressiveness. The experimental results show that our method is effective and significantly outperforms other state-of-the-art methods in both objective and subjective measures. Our method improves the balance of model complexity and performance of the lightweight SISR models to a new level.

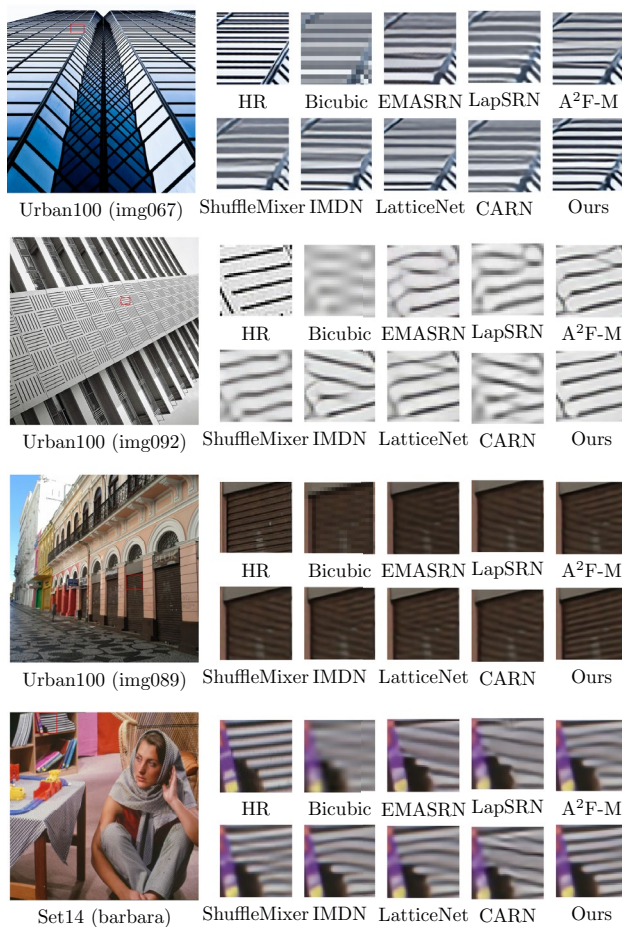


Fig. 3 Visual comparison for 4× SR

Data availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request. The data we used are all publicly available, and the sources of all datasets can be found in the references of the original article.

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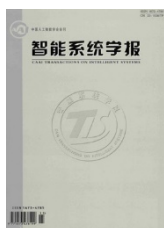
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基于多路特征渐进融合和注意力机制的轻量级图像超分辨率重建

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摘要: 为了进一步探索在计算和存储资源受限设备上应用超分辨率方法的可能性, 本研究聚焦于深度卷积神经网络技术在单图像超分辨率中的应用, 特别是如何在显著增加网络规模的情况下, 提升网络的性能。本文提出了一个新的基于多路特征渐进融合和注意力机制的轻量级单图像超分辨率方法(multi-path feature fusion and attention mechanism, MPFFA)。MPFFA 包括一个多路特征渐进融合块 (multi-path feature progressive fusion, FPF), 可以通过前面的特征, 多路渐进地引导和校准后面特征的学习, 以及包括一个多路特征注意力机制(multi-path feature attention mechanism, FAM), 通过加权拼接多路特征通道, 可以提高特征信息的利用率和特征表达能力。实验结果表明: MPFFA 显著优于当前其他代表性的方法, 在模型复杂度和性能间达到了更好的平衡。本文提出的模型能够更好的应用于计算和资源受限的设备上。

关键词: 图像超分辨率; 卷积神经网络; 特征融合; 注意力机制; 深度学习; 图像还原; 峰值信噪比; 结构相似度
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Lightweight super-resolution reconstruction via progressive multi-path feature fusion and attention mechanism

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Abstract: In order to further explore the possibility of applying super-resolution methods on computing and storage resource-constrained devices, this study focuses on the application of deep convolutional neural network technology in single-image super-resolution, especially how to improve the performance of the network without significantly increasing the network size. In this paper, a novel lightweight SISR method via progressive multi-path feature fusion and attention mechanism (MPFFA) is proposed. MPFFA is effective and significantly outperforms other state-of-the-art methods, thus achieves a better balance between model complexity and performance. The proposed model can be better applied to computing and resource-constrained devices.

Keywords: image super-resolution; convolutional neural network; feature fusion; attention mechanism; deep learning; image restoration; peak signal-to noise ratio; structural similarity

单图像超分辨率(single image super-resolution, SISR)^[1]旨在从给定的低分辨率(low-resolution, LR)图像, 恢复对应的高分辨率(high-resolution, HR)图像, 已经被广泛用于遥感成像、视频监控和医学成像等

计算机视觉领域。然而, SISR 是一个经典的病态问题, 因为一个 LR 图像可以对应多个 HR 图像。

图像超分辨率(super-resolution, SR)技术的研究有很长的历史^[1], 当前, 基于深度卷积神经网络^[2](convolutional neural networks, CNNs)的技术主导了 SISR 技术的研究。直接端到端地学习 LR 和 HR 图像之间的映射函数, 并革新了 SISR 的性能。但是, 基于 CNNs 的 SISR 方法严重依赖于网络的规模, 即网络中参数量、深度(层数)和宽度(通道数)等。要改进 SISR 的性能, 往往需要规模更大的网络, 例如: Lim 等^[3]提出的 EDSR 方法, 有 65 个卷积层, 参数量达到了 43 M; Zhang 等^[4]提出的 RCAN 卷积层数超过了 800 层, 参数量约为 16 M; 用于单幅图像超分辨率的增强型深度残差网络(enhanced deep residual networks for single image super-resolution, EDSR)和非常深的残差通道注意力网络(very deep residual channel attention networks, RCAN)等方法, 虽然有很好的性能, 但是, 需要较高的计算和存储能力, 难以在资源受限的设备上应用。因此, 设计轻量级网络(计算和存储需求较低), 对于当前的 SISR 方法研究是有必要的。但是, 如何在深度 CNN 模型复杂度和 SISR 的性能之间建立更好的平衡, 是一个具有挑战的问题。

多路径学习是一种流行的 SISR 网络架构^[5]。例如: Zhao 等^[6]提出了一种非常轻量级的 SISR 网络, 称为像素注意力网络(pixel attention network, PAN), 该网络仅有 272 K, 却取得了较好的性能。PAN 的 SC-PA (self-calibrated block with pixel attention, SC-PA)块包含 2 个分支: 一个分支包含像素注意力 PA(pixel attention, PA), 该分支主要用于提取更高层的特征信息; 另一个分支使用标准的卷积层, 保持原始的特征信息。2 个分支的输出最后进行通道拼接(concatenation)。Wang 等^[16]的研究表明: 前面层的特征(即辅助特征)有助于当前层特征的学习; Zhao 等^[8]的研究表明: 采用加权的通道拼接, 更符合人脑神经元的行为, 可以提高特征的表达能力。本文主要受 Zhao 等^[6]、Wang 等^[7]和 Zhao 等^[8]的工作启发, 提出了一个基于多路特征渐进融合和注意力机制的轻量级 SISR 网络 (multi-path

feature fusion and attention mechanism network, MPFFA)。其特点在于: 类似于 Zhao 等^[6]提出的 PAN, 核心构件采用了多分支结构(3 个分支), 受 Wang 等^[7]辅助特征学习的启发, 渐进地进行特征提取和融合, 使用前一个分支的特征引导和校准当前分支的特征学习, 并根据 Zhao 等^[8]等加权拼接的研究结果, 提出了一个多路注意力机制, 最优加权各个分支的特征通道, 再进行拼接。MPFFA 网络主干的核心构件是多路特征提取块(multi-path feature extraction, MPFE)。MPFE 包括 2 个子块: 多路特征渐进融合块(multi-path feature progressive fusion, FPF)和多路特征注意力机制(multi-path feature attention mechanism, FAM)。FPF 通过前面的特征, 多路渐进地引导后面特征的提取, 及逐像素预测特征的重要性; FAM 则通过通道注意力, 加权各路特征通道后, 再进行拼接。无论是主观视觉效果, 还是客观度量, 本文提出的 MPFFA 方法, 显著优于当前其他类似的轻量级方法, 把模型复杂度和性能之间的平衡, 提升到一个新的水平。

1 单图像超分辨率网络的相关工作

1.1 轻量级 SISR 方法

Dong 等^[2]第 1 个提出了基于 CNN 的 SISR 方法 SRCNN (super-resolution using convolutional neural network, SRCNN)。之后, 出现了大量基于 CNN 的 SISR 方法^[9]。但是, 基于 CNN 的 SISR 模型, 严重依赖于网络规模。通常, 网络规模越大, 表达能力越强, 性能也越好, 然而, 更大的网络往往会急剧增加参数量和计算量, 这也限制了它们在计算和存储资源受限的设备上的使用。为了解决这一问题, 出现了大量基于 CNN 的轻量级 SISR 方法。递归是一种流行的轻量级 SISR 网络结构^[10-12]。递归的结构, 虽然可以减少参数量, 但是, 不能减少计算量。为了进一步解决这一问题, 1×1 卷积^[5-6, 13-14]、分组卷积^[15]、深度可分离卷积^[15]和自校准卷积^[6]等也在轻量级 SISR 网络架构中广泛使用。本文中, 采用了多路径学习和多路注意力机制, 设计了一种轻量的构建块, 减少了参数量和计算量, 提高了 SISR 性能和计算效

率。

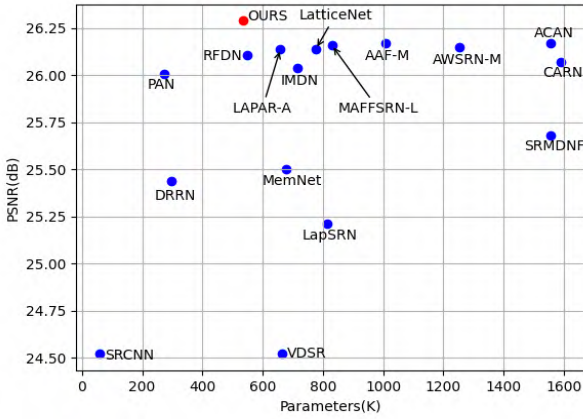


图 1 各个先进 SISR 方法的 4 倍 SR 模型, 在 Urban100^[15]数据集上的参数数量和平均 PSNR 结果

Fig. 1 Average PSNRs and numbers of parameters for $\times 4$ SR models for each state of the art SISR method on the Urban100 dataset

1.2 多路径学习

多路径学习是 CNNs 架构的一个重要形式, 残差连接^[16]和稠密连接^[17], 可能是 SISR 网络架构中最常见的多路径学习。残差连接和稠密连接能够增强特征信息的传播, 但是, 也会造成大量的特征信息冗余。如何改进多路径学习, 提高特征信息的利用率和表达能力, 持续受到研究人员的重视^[13]。Wang 等^[7]提出的注意力辅助特征块 A^2F (attentive auxiliary features, A^2F), 包含 2 个主要分支: 一个分支与前面模块的输出特征进行通道拼接和压缩, 再使用通道注意力机制, 学习各个通道的重要性; 另一个分支执行卷积运算。Zhao 等^[6]提出的 PAN, 基本组成块是自校准块 SC-PA。SC-PA 块包含 2 个分支: 一个分支使用 PA, 用于提取更高层的特征信息; 另一个分支使用标准的卷积层, 保持原始的特征信息。受到 PAN 和 A^2F 的启发, 本文采用了 3 条支路, 并且可以通过前面的特征, 多路渐进地引导后面特征的提取。

1.3 注意力机制

注意力机制^[15]通常是指人类的视觉系统, 能够自动地聚焦于显著的区域。卷积神经网络网络架构中的注意力机制与其类似, 是指能够聚焦于输入的特定部分, 提高特征的利用率和表达能力^[4,18]。Zhang 等^[19]提出的一个多上下文注意力块 MCAB (multi-context attentive block, MCAB), 还可以同时关注通道维度和空间维度之间的依赖关系。本文提出的多路特征注意力机制, 通过加权拼接多路通道特征, 进一步增强特征表达能力。

2 本文方法

本文提出的基于多路特征渐进融合和注意力机制的轻量级 SISR 网络(MPFFA)架构, 参见图 2(a)。MPFFA 主要包括 4 个部分: 一个 3×3 的卷积层、一个深层特征提取块(deep feature extraction block, DFEB)、一个分层特征融合注意力块(hierarchical feature fusion attention block, HFFAB)和一个上采样块(Upsampler)。 3×3 的卷积层用于提取浅层特征。Upsampler 使用了 PAN^[6]的上采样块。HFFAB 主要使用了 TSAN^[19]中的多上下文注意力块 MCAB。为了减小计算量, 对输入到 MCAB 的特征通道, 用 1×1 的卷积进行了 4 倍的压缩和融合; 对 MCAB 的输出特征通道, 再用 1×1 的卷积进行了 4 倍的扩张。本文工作主要是 DFEB, 包括 3 个特征提取组(feature extraction group, FEG)。每个 FEG 包含 6 个多路特征提取块(MPFE), 参见图 2(b)。FEG 提取的分层特征进行拼接, 并经 HFFAB 注意力块进行融合, 以更高效地聚焦和利用信息丰富的上下文特征。组成 FEG 的 MPFE 是本文模型的核心构件。每个 MPFE 包括一个多路特征渐进融合块(FPF)和一个多路特征注意力机制(FAM), 参见图 2(d)。FPF 和 FAM 的详细介绍, 参见下文 2.1 和 2.2。MPFFA 架构也包含一个全局残差连接, LR 图像通过双三次插值, 放大到目标 SR 图像的大小。下面详细介绍 MPFFA 的各个部分。

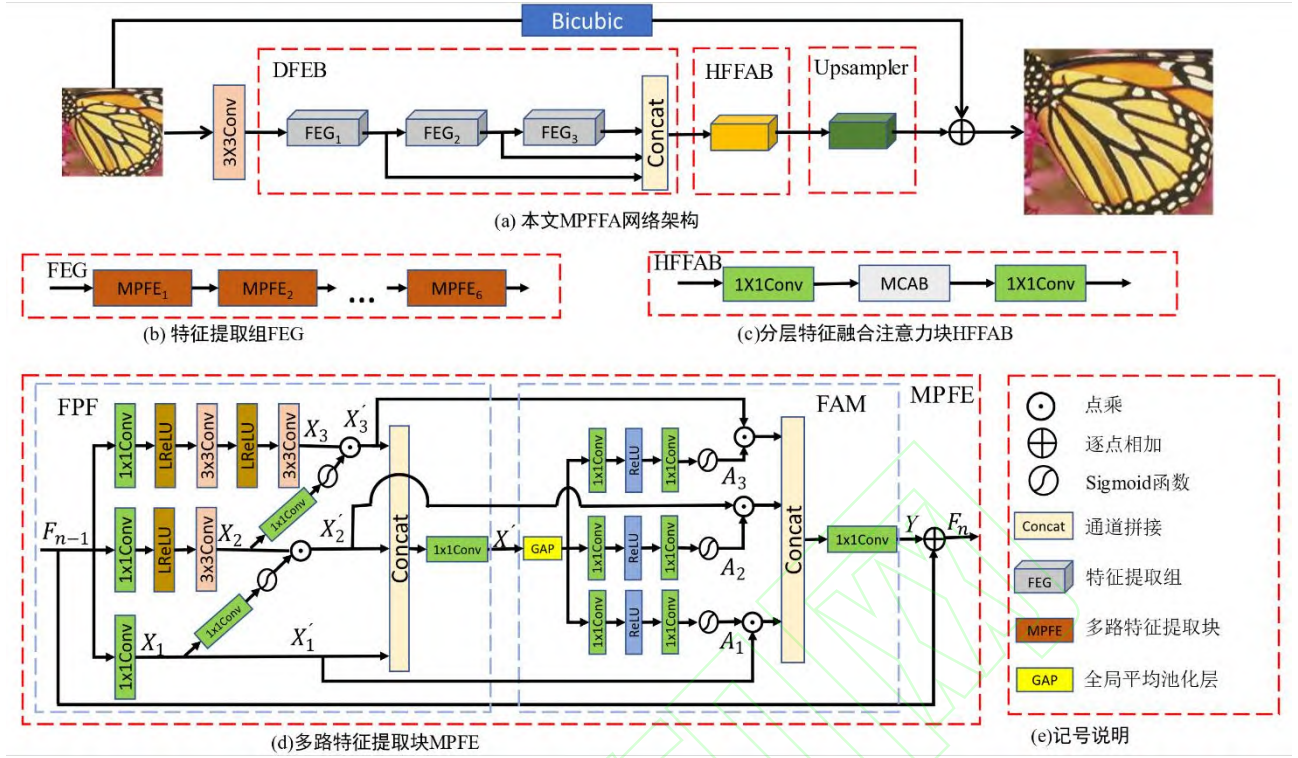


图2 本文模型架构图

Fig.2 The model architecture of this paper

假定输入的 LR 图像为 I_{LR} , I_{LR} 首先通过一个 3×3 的卷积层, 提取浅层特征

$$F_0 = \text{Conv}_{3 \times 3}(I_{LR})$$

式中: $\text{Conv}_{3 \times 3}(\cdot)$ 为 3×3 的卷积, F_0 为其输出的特征信息。 F_0 再输入到 DFEB, 进行深层特征提取。 DFEB 包含 3 个 FEG, 每个 FEG 又包含 6 个 MPFE。 所以, F_0 共经过 18 个 MPFE, 每个 MPFE 的输出可以表示为

$$F_n = H_n(F_{n-1}), n=1, 2, \dots, 18 \quad (1)$$

式中: $H_n(\cdot)$ 为第 n 个 MPFE 函数, F_n 为其输出特征信息。 每个 FEG 的输出分别为 F_6 、 F_{12} 和 F_{18} , 进行通道拼接

$$F_{M0} = [F_6, F_{12}, F_{18}]$$

式中 $[\cdot]$ 为特征通道拼接, F_{M0} 为拼接后的特征, 也是 DFEB 块的输出特征, 再输入到 HFFAB

$$F_M = H_{\text{HFFAB}}(F_{M0}) \quad (2)$$

式中: $H_{\text{HFFAB}}(\cdot)$ 为 HFFAB 函数, F_M 为其输出的特征信息。 F_M 再输入上采样块

$$I_{SR} = H_{\text{Up}}(F_M) + H_{\text{Bic}}(I_{LR})$$

式中 $H_{\text{Up}}(\cdot)$ 为上采样块 Upsampler 函数; $H_{\text{Bic}}(\cdot)$ 为

传统的双三次插值函数, 叠加到残差 $H_{\text{Up}}(\cdot)$ 的输出, 得到最终输出的目标 SR 图像 I_{SR} 。 当前 SISR 方法的上采样块, 大多数采用亚像素的卷积^[3]或转置的卷积^[9], 本文使用了 PAN^[6]的上采样块。

2.1 多路特征渐进融合块(FPF)

FPF 主要是受 Zhao 等^[6]和 Wang 等^[7]工作的激发, 进行多路(本文使用 3 路)渐进特征提取和融合, 并用前面的特征引导和校准后面的特征学习。 第 1 条支路仅包含一个 1×1 的卷积层; 第 2 条支路包含了一个 1×1 的卷积层和一个 3×3 的卷积层, 2 个卷积层之间包含一个 LReLU^[15]非线性激活函数, 第一条支路的特征通过 1×1 的卷积和 Sigmoid 函数, 引导和校准该支路特征; 第 3 条支路包含一个 1×1 的卷积层和 2 个 3×3 的卷积层, 2 个卷积层之间包含一个 LReLU 非线性激活函数, 第 2 条支路的特征再通过 1×1 的卷积和 Sigmoid 函数, 引导和校准该支路特征。 考虑到 F_{n-1} 和 F_n 分别是第 n 个 MPFE 的输入和输出, 参见式(1)。 3 条支路校准前的特征输出分别为(忽略了 LReLU 非线性激活函数)

$$X_1 = \text{Conv}_{1 \times 1}(F_{n-1})$$

$$X_2 = \text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F_{n-1}))$$

$$X_3 = \text{Conv}_{3 \times 3}(\text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F_{n-1})))$$

式中: $\text{Conv}_{1 \times 1}(\cdot)$ 为 1×1 的卷积, $\text{Conv}_{3 \times 3}(\cdot)$ 为 3×3 的卷积, 3 条支路校准后的特征输出分别为

$$X'_1 = X_1$$

$$X'_2 = X_2 \odot (\sigma(\text{Conv}_{1 \times 1}(X_1)))$$

$$X'_3 = X_3 \odot (\sigma(\text{Conv}_{1 \times 1}(X_2)))$$

式中: $\sigma(\cdot)$ 为 Sigmoid 函数, $\odot$ 为对应元素相乘。然后, 3 条支路校准后的输出特征通道进行拼接, 并用 1×1 的卷积进行通道压缩和融合。

$$X' = \text{Conv}_{1 \times 1}([X'_1, X'_2, X'_3])$$

式中: $\text{Conv}_{1 \times 1}(\cdot)$ 为 1×1 的卷积, $[\cdot]$ 为特征通道拼接。

2.2 多路特征注意力机制(FAM)

FAM 主要是受 Zhao 等^[8]工作的激发, 加权的特征通道拼接, 可以提高特征的表达能力。本文算法采用注意力机制, 可以最优加权特征通道, 并进行拼接。第 n 个 FPF 块的输出特征 X' , 通过全局平均池化(GAP), 生成初始权重向量。然后, 各个支路通过 2 个 1×1 的卷积(2 个 1×1 的卷积之间包含 1 个 ReLU^[15]非线性激活函数), 进行通道的压缩和扩张, 再通过 1 个 Sigmoid 函数, 生成各个支路特征通道的权重向量。

假定 FPF 块的特征拼接输出 $X' = [x_1, \dots, x_c, \dots, x_C] \in \mathbb{R}^{H \times W \times C}$, GAP 运算可表示为

$$z_c = H_{\text{GAP}}(x_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j)$$

式中 $H_{\text{GAP}}(\cdot)$ 为 GAP 函数, $Z = [z_1, \dots, z_c, \dots, z_C] \in \mathbb{R}^C$, 3 个支路特征各个通道的权重可表示为(忽略了 ReLU 非线性激活函数)

$$A_k = \sigma(\text{Conv}_{1 \times 1}(\text{Conv}_{1 \times 1}(Z))), k = 1, 2, 3$$

式中 $\text{Conv}_{1 \times 1}(\cdot)$ 为 1×1 的卷积, $\sigma(\cdot)$ 为 Sigmoid 函数。第 n 个 FAM 的残差输出 Y , 可表示为

$$Y = \text{Conv}_{1 \times 1}([A_1 \odot X'_1, A_2 \odot X'_2, A_3 \odot X'_3])$$

式中: $\text{Conv}_{1 \times 1}(\cdot)$ 为 1×1 的卷积, $X'_k (k=1, 2, 3)$ 为第 n 个 FPF 块的 3 条支路输出的特征, $[\cdot]$ 为特征通道拼接, $\odot$ 为权重向量元素与对应特征通道相乘。由于使用了残差连接, 第 n 个 FAM 的输出特征 F_n ,

需要再加上第 n 个 FPF 块的输入 F_{n-1} , 即

$$F_n = Y + F_{n-1}$$

2.3 分层特征融合注意力模块(HFFAB)

MCAB^[19]可聚焦信息丰富的上下文特征。HFFAB 只是在多上下文注意力块(MCAB)前后各增加一个 1×1 的卷积层。前一个 1×1 的卷积层压缩特征通道, 以减少计算量; 后一个 1×1 的卷积层再扩张特征通道。HFFAB 块的运算, 参见式(2), 可进一步表示为

$$F_M = \text{Conv}_{1 \times 1}(H_{\text{MCAB}}(\text{Conv}_{1 \times 1}(F_{M0})))$$

3 训练 MPFFA 实验细节

3.1 实验设置

训练阶段, 本文采用流行的 DIV2K^[20]数据集, 作为训练和验证数据集。前 800 张图像(001-800)用于训练, 10 张图像(821-830)用于验证, 标记为 DIV2K_val10。为了制作这些训练对, 我们首先从高质量的 HR 图像开始, 然后应用双三次插值法进行下采样, 以生成对应的 LR 图像。除了下采样, 为了提高模型的泛化能力, 还需要对训练数据集进行数据增强。这包括随机地对图像进行 90° 、 180° 、 270° 的随机旋转以及水平翻转以对图像进行数据增强。首先训练好的 $\times 2$ 模型, 然后通过预训练的 $\times 2$ 模型, 训练 $\times 3$ 和 $\times 4$ 的模型。测试中, 使用了 Set5、Set14、B100、Urban100 和 Manga109 共 5 个标准的测试数据集^[6]。为了公正的比较, 与之前的方法一样, 在 YCbCr 空间^[17]的亮度(Y)通道上, 计算峰值信噪比(peak signal to noise ratio, PSNR)和结构相似性(structural similarity index, SSIM)^[3]度量。同时本文也计算了学习的感知图像块相似性(learned perceptual image patch similarity, LPIPS)^[23]度量。

模型训练中, 每批次随机选取 32 个 48×48 的图像块, 使用 Adam^[22]优化器, $\beta_1 = 0.9$ 、 $\beta_2 = 0.999$ 和 $\varepsilon = 10^{-8}$ 。本文模型训练了 1200 个迭代周期, 初始学习率设置为 2×10^{-4} , 每 200 个迭代周期衰减一半。使用的损失函数是 L_1 。通过 PyTorch^[22]框架, 一个 NVIDIA 2080Ti GPU 实现模型, 并进行模型的训练和测试。

3.2 消融实验

为了验证 FPF、FAM, 以及 HFFAB 的有效性, 进行了 4 组对比实验。第 1 个实验, 不包含 FAM 和

HFFAB, FPF 块仅包含 3 个主支路, 不包含前面的特征对后面的特征的校准, 即不包含 2 个支路之间的 1×1 的卷积和 Sigmoid 函数, 这个模型称为 MPFFA_0。第 2 个实验, 在 MPFFA_0 基础上增加包含前面的特征对后面的特征的校准, 即完整包含 FPF, 仍不包含 FAM 和 HFFAB, 这个模型称为 MPFFA_1。第 3 个实验, 是在 MPFFA_1 的基础上, 增加 HFFAB, 这个模型称为 MPFFA_2。第 4 个实验, 是在 MPFFA_2 的基础上, 再增加 FAM, 即完整的 MPFFA 网络模型。4 个模型在 DIV2K_val10 验证集上, 3 倍 SR 的 PSNR, 参见表 1。同时也给

出了各个模型的数量和计算量(即 GFLOPs^[15])和 LPIPS 度量^[21]结果, 以供参考, 其中 LPIPS 值越小越好。可以看到: 模型 MPFFA_0 的 PSNR 仅 29.49 dB; 模型 MPFFA_1 的 PSNR 增加了 0.11 dB, 表明多路特征渐进融合, 前面特征辅助后面特征的学习, 是很有效的; 相较于模型 MPFFA_1, 模型 MPFFA_2 仅增加了 15 K 的参数量和 1.4 G 的计算量, 使 PSNR 值增加了 0.03 dB。最终的模型 MPFFA, 相较于模型 MPFFA_2, PSNR 又增加了 0.06 dB, 表明对各个支路特征通道进行加权拼接的 FAM 块, 是有效的。

表 1 网络架构不同变种, 在验证集 DIV2K_val10 上, 3 倍 SR 的 PSNR、参数量、计算量和 LPIPS

Table 1 Average PSNRs, number of parameters, computational cost and LPIPSs for $\times 3$ SR on validation dataset DIV2K_val10 for different variants of network architecture.

模型	FPF 校准部分	FPF 三个主分支	HFFAB	FAM	PSNR (dB)	参数量	计算量	LPIPS
MPFFA_0	×	√	×	×	29.49	443K	59.4G	0.1861
MPFFA_1	√	√	×	×	29.60	465K	61.6G	0.1836
MPFFA_2	√	√	√	×	29.63	480K	63.0G	0.1826
MPFFA	√	√	√	√	29.69	523K	66.2G	0.1813

3.3 特征提取组(FEG)实验

本文模型包含 3 个 FEG, 每组内包含相同个数 MPFE。MPFE 的个数选择, 要综合考虑参数量、计算量, 以及每个 FEG 输出的不同深度特征拼接后, 对性能的影响。本文在 DIV2K_val10 验证集上, 进行了 3 组实验。每组实验中, 每个 FEG 中 MPFE 个数分别设置为 5、6 和 7。表 2 给出了模型在不同 MPFE 个数下的 3 倍超分辨率 PSNR 结果。通过结果分析, 可以明显观察到随着 MPFE 个数的增加, 模型在 3 倍超分辨率任务上的性能得到了提升。这表明 MPFE 能够有效地丰富模型对图像的理解, 从而更好地重建高分辨率图像。然而, 需要注意的是, 随着 MPFE 个数的增加, 模型的参数量也相应增长。这可能会导致模型变得更加复杂, 需要更多的计算资源和存储空间。因此, 在实际应用中, 需要权衡性能提升和参数量增加之间的关系, 找到一个合理的平衡点, 以确保模型既有较好的性能, 又能保持

较低的计算成本。相较于 5 个 MPFE, 6 个 MPFE 的 PSNR 增加了 0.14dB; 相较于 7 个 MPFE, 6 个 MPFE 的 PSNR 仅减少了 0.05dB。7 个 MPFE 的 PSNR 增幅显著减缓, 而参数量的增加是一样的, 都是 78 K。为了保持网络模型足够轻量, MPFE 个数取 6 (本文中的设置), 可能是一个合理的折中。

表 2 在 DIV2K_val10 上 3 倍 SR 实验, 每个 FEG 中不同 MPFE 个数, 对模型平均 PSNR 性能和参数量的影响

Table 2 Average PSNRs and numbers of parameters for $\times 3$ SR experiments on DIV2K_val10 for different number of MPFEs in each FEG

MPFE/个	PSNR /dB	参数量/K
5	29.55	445K
6	29.69	523K
7	29.74	601K

3.4 多路特征注意力机制(FAM)实验

3.4.1 多支路的效果

为了进一步洞察 FAM 中多支路的效果, 本文把 FAM 块改成了 1 条支路, 在 DIV2K_val10 验证集上进行实验。3 倍 SR 的 PSNR 结果为 29.56 dB, 这比 FAM 包含 3 条支路的 MPFFA 模型的 PSNR 低 0.13 dB。一个有趣的结果是: FAM 仅包含一条支路的模型, 比 MPFFA 中去掉 FAM 块的模型(MPFFA_2)的 PSNR, 还要低 0.07 dB。

3.4.2 参数共享的效果

参数共享, 可以减少参数量。如果 FAM 中各支路参数共享, 在验证集 DIV2K_val10 上, 3 倍 SR 的 PSNR 结果为 29.65 dB, 比 MPFFA 中各支路参数不共享的情形, PSNR 低 0.04 dB。

3.5 与先进方法比较

为了检验本文提出的 MPFFA 方法的有效性, 将其与其他前沿技术进行了的客观指标的对比, 以

及从主观视角出发的视觉效果评估的比较。其他代表性的方法包括 SRCNN^[2]、FSRC-NN^[9]、VDSR^[16]、DRCN^[10]、MemNet^[11]、IMDN^[13]、RFDN^[14]、PAN^[6]、A²F-M^[7]、LapSRN^[23]、DRRN^[24]、SRMDNF^[25]、CARN^[26]、LatticeNet^[27]、MAFFSRN-L^[28]、ACAN^[29]和 LAPAR-A^[30]。在 5 个标准测试数据集上, 2、3 和 4 倍 SR 的 PSNR/SSIM、参数量和计算量(即 GFLOPs^[15])结果, 如表 3 所示, 视觉效果如图 3 所示。除 LatticeNet 和 PAN 以外, 其他方法的 PSNR/SSIM 结果均来自作者的原论文。LatticeNet 的结果, 源自运行文中提供的训练模型^[27]。由于 PAN 使用了 DIV2K 和 Flickr2K^[6]2 个训练数据集, 为了公平比较, 使用了文中提供的代码, 在 DIV2K 训练集上, 重新进行了训练, 所以, 测试结果与原论文上的结果略有不同。

表 3 各种先进 SISR 方法的平均 PSNR/SSIM、参数量和计算量结果

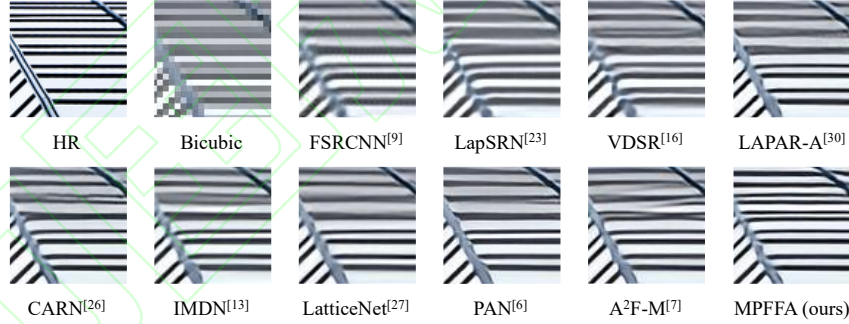
Table 3 Average PSNRs/SSIMs, number of parameters and computational cost of various state-of-the-art SISR methods

放大倍数	模型	参数量/K	计算量/G	Set5	Set14	B100	Urban100	Manga109
				PSNR/SSIM	PSNR/SSIM	PSNR/SSIM	PSNR/SSIM	PSNR/SSIM
×2	SRCNN	57	52.7	36.66/0.9524	32.42/0.9063	31.36/0.8879	29.50/0.8946	35.74/0.9661
	FSRCNN	12	6	37.00/0.9558	32.63/0.9088	31.53/0.8920	29.88/0.9020	36.67/0.9694
	VDSR	665	612.6	37.53/0.9587	33.03/0.9124	31.90/0.8960	30.76/0.9140	37.22/0.9729
	DRCN	1774	17974	37.63/0.9588	33.04/0.9118	31.85/0.8942	30.75/0.9133	37.63/0.9723
	LapSRN	813	29.9	37.52/0.9590	33.08/0.9130	31.80/0.8950	30.41/0.9100	37.27/0.9740
	DRRN	297	6796.9	37.74/0.9591	33.23/0.9136	32.05/0.8973	31.23/0.9188	37.92/0.9760
	MemNet	677	2662.4	37.78/0.9597	33.28/0.9142	32.08/0.8978	31.31/0.9195	37.72/0.9740
	SRMDNF	1513	347.7	37.79/0.9600	33.32/0.9150	32.05/0.8980	31.33/0.9200	38.07/0.9761
	CARN	1592	222.8	37.76/0.9590	33.52/0.9166	32.09/0.8978	31.92/0.9256	38.36/0.9765
	IMDN	694	158.8	38.00/0.9605	33.63/0.9177	32.19/0.8996	32.17/0.9283	38.88/0.9774
	LAPAR-A	548	171	38.01/0.9605	33.62/0.9183	32.19/0.8999	32.10/0.9283	38.67/0.9772
	RFDN	534	123.0	38.05/0.9606	33.68/0.9184	32.16/0.8994	32.12/0.9278	38.88/0.9773
	A ² F-M	999	224.2	38.04/0.9607	33.67/0.9184	32.18/0.8996	32.27/0.9294	38.87/0.9774
	ACAN	800	2108	38.10/0.9608	33.60/0.9177	32.21/0.9001	32.29/0.9297	38.81/0.9773
	LatticeNet	756	169.5	38.06/0.9607	33.70/0.9187	32.20/0.8999	32.25/0.9288	38.94/0.9774
	AWSRN-M	1063	244.1	38.04/0.9605	33.66/0.9181	32.21/0.9000	32.23/0.9294	38.66/0.9772
	PAN	261	70.5	37.99/0.9603	33.53/0.9174	32.14/0.8992	31.93/0.9263	38.37/0.9766
	MAFFSRN-L	790	154.4	38.07/0.9607	33.59/0.9177	32.23/0.9005	32.38/0.9308	-/-
	MPFFA	523	129.4	38.09/0.9607	33.70/0.9190	32.21/0.9000	32.34/0.9296	38.98/0.9767
×3	SRCNN	57	52.7	32.75/0.9090	29.28/0.8209	28.41/0.7863	26.24/0.7989	30.59/0.9107
	FSRCNN	12	4.6	33.16/0.9104	29.43/0.8242	28.53/0.7910	26.43/0.8080	30.98/0.9212
	VDSR	665	612.6	33.66/0.9213	29.77/0.8314	28.82/0.7976	27.14/0.8279	32.01/0.9310
	DRCN	1774	17974	33.82/0.9226	29.76/0.8311	28.80/0.7963	27.15/0.8276	32.31/0.9328
	DRRN	297	67969.0	34.03/0.9244	29.96/0.8349	28.95/0.8004	27.53/0.8378	32.74/0.9390
	MemNet	677	2662.4	34.09/0.9248	30.00/0.8350	28.96/0.8001	27.56/0.8376	32.51/0.9369
	SRMDNF	1530	156.3	34.12/0.9250	30.04/0.8370	28.97/0.8030	27.57/0.8400	33.00/0.9403
	CARN	1592	118.8	34.29/0.9255	30.29/0.8407	29.06/0.8034	27.38/0.8404	33.50/0.9440
	IMDN	703	71.5	34.36/0.9270	30.32/0.8417	29.09/0.8046	28.17/0.8519	33.61/0.9445
	LAPAR-A	594	114	34.36/0.9267	30.34/0.8421	29.11/0.8054	28.15/0.8523	33.51/0.9441

	RFDN	541	55.4	34.41/0.9273	30.34/0.8420	29.09/0.8050	28.21/0.8525	33.67/0.9449
	A2F-M	1003	100	34.50/0.9278	30.39/0.8427	29.11/0.8054	28.28/0.8546	33.66/0.9453
	ACAN	1115	1051.7	34.46/0.9277	30.39/0.8435	29.11/0.8055	28.28/0.8550	33.61/0.9447
	LatticeNet	765	76.3	34.40/0.9272	30.32/0.8416	29.10/0.8049	28.19/0.8513	33.63/0.9442
	AWSRN-M	1143	116.6	34.42/0.9275	30.32/0.8419	29.13/0.8059	28.26/0.8545	33.64/0.9450
	PAN	261	39.0	34.30/0.9266	30.30/0.8416	29.06/0.8042	28.02/0.8493	33.34/0.9435
	MAFFSRN-L	807	68.5	34.45/0.9277	30.40/0.8432	29.13/0.8061	28.26/0.8552	-/-
	MPFFA	523	66.2	34.51/0.9283	30.37/0.8424	29.13/0.8058	28.35/0.8550	33.85/0.9461
×4	SRCNN	57	52.7	30.48/0.8628	27.49/0.7503	26.90/0.7101	24.52/0.7221	27.66/0.8505
	FSRCNN	12	4.6	30.71/0.8657	27.59/0.7535	26.98/0.7150	24.62/0.7280	27.90/0.8517
	VDSR	665	612.6	31.35/0.8838	28.01/0.7674	27.29/0.7251	25.18/0.7524	28.83/0.8809
	DRCN	1774	17974	31.53/0.8854	28.02/0.7670	27.23/0.7233	25.14/0.7510	28.98/0.8816
	LapSRN	813	149.4	31.54/0.8850	28.19/0.7720	27.32/0.7280	25.21/0.7560	29.09/0.8845
	DRRN	1774	17974.3	31.53/0.8854	28.02/0.7670	27.23/0.7233	25.14/0.7510	28.98/0.8816
	MemNet	677	2662.4	31.74/0.8893	28.26/0.7723	27.40/0.7281	25.50/0.7630	29.42/0.8942
	SRMDNF	1555	89.3	31.96/0.8930	28.35/0.7770	27.49/0.7340	25.68/0.7730	30.09/0.9024
	CARN	1592	90.9	32.13/0.8937	28.60/0.7806	27.58/0.7349	26.07/0.7837	30.47/0.9084
	IMDN	715	40.9	32.21/0.8948	28.58/0.7811	27.56/0.7353	26.04/0.7838	30.45/0.9075
	LAPAR-A	659	94	32.15/0.8944	28.61/0.7818	27.61/0.7366	26.14/0.7871	30.42/0.9074
	RFDN	550	31.6	32.24/0.8952	28.61/0.7819	27.57/0.7360	26.11/0.7858	<u>30.58/0.9089</u>
	A ² F-M	1010	56.7	<u>32.28/0.8955</u>	28.62/0.7828	27.58/0.7364	<u>26.17/0.7892</u>	<u>30.57/0.9100</u>
	ACAN	1556	616.5	32.24/0.8955	28.62/0.7824	27.59/0.7366	26.17/0.7891	30.53/0.9086
	LatticeNet	777	43.6	32.18/0.8943	28.61/0.7812	27.57/0.7355	26.14/0.7844	30.54/0.9075
	AWSRN-M	1254	72	32.21/0.8954	<u>28.65/0.7832</u>	<u>27.60/0.7368</u>	26.15/0.7884	30.56/0.9093
	PAN	272	28.2	32.06/0.8939	28.56/0.7813	27.55/0.7352	26.01/0.7830	30.33/0.9069
	MAFFSRN-L	830	38.6	32.20/0.8953	28.62/0.7822	27.59/0.7370	26.16/0.7887	-/-
	MPFFA	534	43.0	32.36/0.8968	28.69/0.7835	27.61/0.7372	26.29/0.7902	30.71/0.9115



Urban100 (img067)



Urban100 (img089)

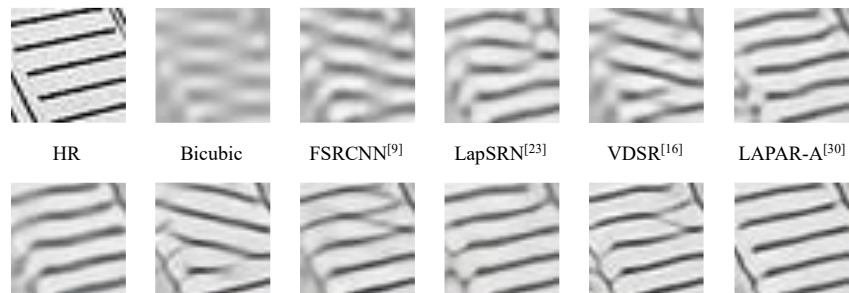
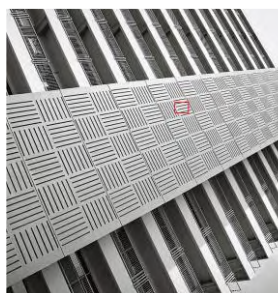
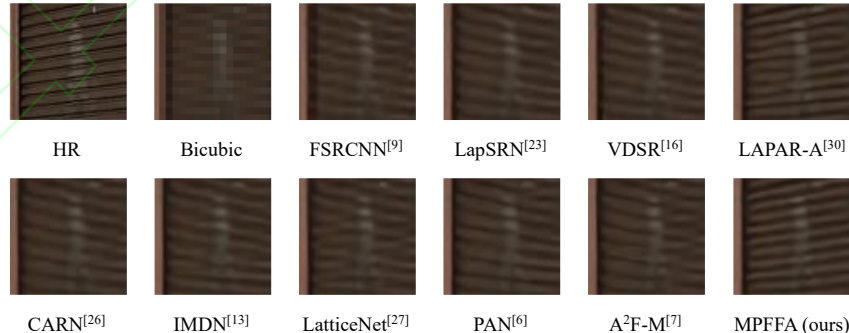




图3 4倍SR结果的视觉效果比较

Fig. 3 Visual comparison for $\times 4$ SR

在 Set5、Urban100 和 Manga109 标准测试数据集上, 与其他 5 个代表方法: CARN^[26]、IMDN^[13]、LatticeNet^[27]、LAPAR-A^[30]和 PAN^[6], 4 倍 SR 的 LPIPS 结果, 如表 4 所示, 值越小越好。

表4 不同SISR方法4倍SR的LPIPS值

Table 4 LPIPSs of $\times 4$ SR for different SISR methods

模型	参数量/K	计算量/G	Set5	Urban100	Manga109
CARN	1592	90.9	0.1761	0.2363	---
IMDN	715	40.9	0.1743	0.2350	0.1330
LatticeNet	777	43.6	0.1769	0.2423	0.1113
LAPAR-A	659	94.0	0.1759	0.2317	0.1101
PAN	272	28.2	0.1741	0.2348	0.1107
MPFFA	534	43.0	0.1727	0.2265	0.1071

注: 加粗字体为每列最优值。

1) 定量客观比较。从表 3 可以看出, 本文方法的 PSNR/SSIM 结果显著优于其他比较的方法。例如, LAPAR-A 和 RFDN 方法, 比本文的 MPFFA 模型略大, 但是, PSNR 的结果一致优于这 2 个方法。 $\times 2$ 、 $\times 3$ 、 $\times 4$ 的 PSNR, 在各个数据集上, 超出 LAPAR-A 最大值分别为 0.31、0.34 和 0.29 dB; 超出 RFDN 最大值分别为 0.22、0.18 和 0.18 dB。A²F-M 方法的参数量, 大约是 MPFFA 参数量的 2 倍, 但是, 在 Manga109 数据集上 $\times 2$ 、 $\times 3$ 、 $\times 4$ 的 PSNR 结果, 比 MPFFA 的结果分别低 0.11、0.19 和 0.14 dB。ACAN 方法的 $\times 4$ 参数量, 大约是 MPFFA 参数量的 3 倍, 但是, 在 Manga109 数据集上的 PSNR 结果, 比 MPFFA 的结果低 0.18 dB, 而且计算量比 MPFFA 大约高出了 26 倍。从表 4 可以看出, 本文方法的 LPIPS 结果也均优于其他比较的方

法。

2) 主观效果比较。图 3 给出了本文方法与其他先进方法在 $\times 4$ 时重建的 SR 图像结果。本文方法的结果比其他方法的结果更好, 是显著可视的。以 Urban100 数据集中图像 img067 为例, Bicubic、FSRCNN、CARN、LatticeNet 和 A²F-M 等方法恢复的楼房条纹, 均有不同程度的模糊和失真, 甚至 A²F-M 方法恢复的条纹也发生了部分扭曲, 本文方法恢复出的条纹很接近于原 HR 图像。Urban100 数据集中图像 img089, 除 LAPAR-A 方法和本文网络模型之外, 其他方法恢复的条纹, 基本上过度模糊失真和条纹方向错误。虽然 LAPAR-A 方法正确地恢复了条纹的方向, 但是, 仍然有明显的模糊, 本文方法的结果更加清晰, 更接近于原 HR 图像。Urban100 数据集中图像 img092, 除 LAPAR-A、PAN 方法和本文网络模型之外, 其他方法恢复的条纹, 基本上存在过度模糊失真和条纹方向错误的现象。虽然 LAPAR-A、PAN 正确地恢复了条纹方向, 但是仍然有明显的模糊, 本文方法的结果更加清晰, 与原 HR 图像更加接近。而对于 Set14 数据集中的图像 barbara, 只有本文方法对纹理的重建方向正确, 而且视觉效果更加清晰且不失真。

图 4 给出了本文方法与其他先进方法在 $\times 3$ 时重建的 SR 图像结果。本文方法的结果比其他方法的结果更好是显著可视的。以 Urban100 数据集中图像 img011 为例, Bicubic、FSRCNN、CARN、LatticeNet 和 A²F-M 等方法恢复的纹理, 均有不同程度的模糊和失真, 本文方法的重建效果很接近于原 HR 图像。Urban100 数据集中图像 img024 也类似, 相较于其他几种方法, 本文方法的重建效果比它们更接近于 HR 图像。

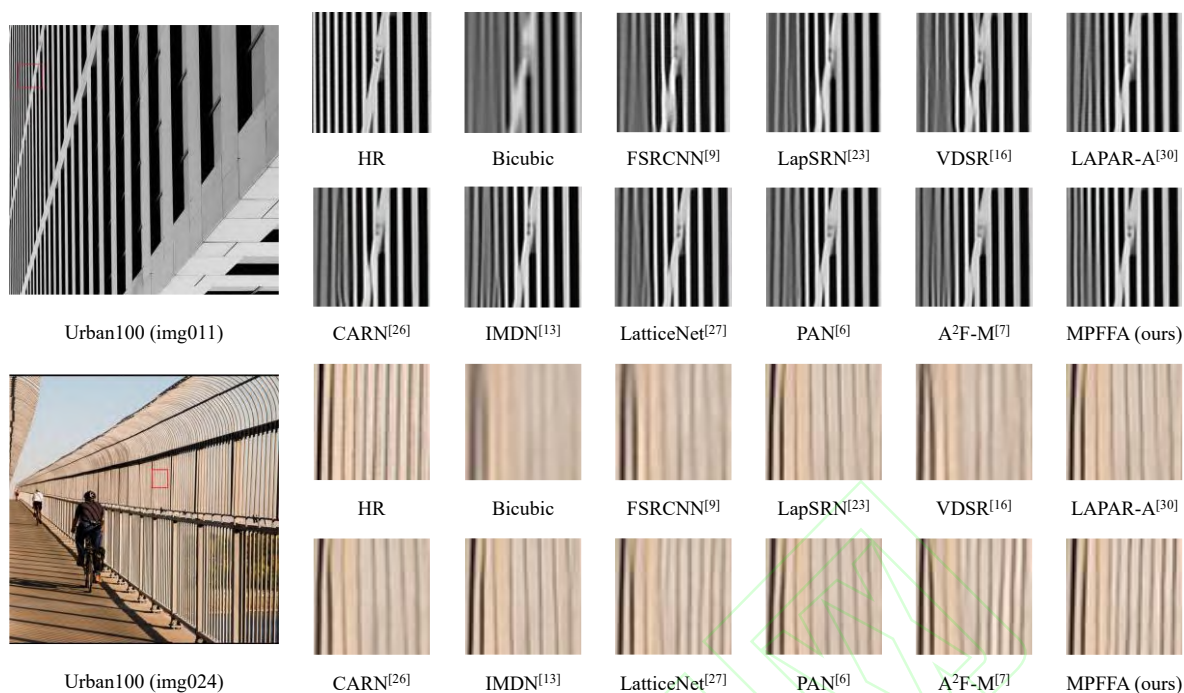


图4 3倍SR结果的视觉效果比较

Fig. 4 Visual comparison for $\times 3$ SR

4 结束语

本文提出了一个基于多路特征渐进融合和注意力机制的轻量级图像超分辨率网络。本文主要工作是提出了一个 FPF 和一个 FAM, FPF 和 FAM 块配对使用。FPF 可以利用前面的特征, 多路渐进地引导和辅助后面特征的学习; FAM 则可以把多路特征通道, 加权拼接, 提高特征信息的利用率和表达能力。实验结果表明: 本文方法是有效的, 无论是客观度量还是主观度量, 都明显优于其他先进的方法, 把轻量级 SISR 的复杂度和性能平衡, 提高到了一个新的水平。未来仍考虑进一步优化本文网络结构和损失函数, 继续改进其性能。

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基于全局依赖 Transformer 的图像超分辨率网络

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摘要: 目前, 基于深度学习的图像超分辨率网络主要由卷积实现。相较于传统的卷积神经网络, Transformer 在图像超分辨率任务中的主要优势是它的长距离依赖建模能力。然而大多数基于 Transformer 的图像超分辨率模型在参数量小、网络层数少的情况下无法建立全局依赖, 限制了模型的性能。为了在超分辨率网络中建立全局依赖, 提出了基于全局依赖 Transformer 的图像超分辨率网络, 其主要组成部分为残差方形轴向窗口块, 其内部的轴向窗口 Transformer 残差层利用轴向窗口和自注意力, 可以使每个像素与整个特征图建立起全局依赖。此外, 目前大多数图像超分辨率模型的超分辨率图像重建模块都由卷积组成, 为了可以动态整合提取到的特征信息, 把 Transformer 与卷积进行结合, 共同重建超分辨率图像。通过实验表明, 基于全局依赖 Transformer 的图像超分辨率网络在各个倍数的超分辨率任务中, 与其他先进的模型相比, 在大尺寸数据集上的峰值信噪比指标上有超过 0.3dB 的提升。

关键词: 图像超分辨率; Transformer; 自注意力; 全局依赖; 轴向窗口

中图分类号: TP391.4; TP183

文献标志码: A

Image super-resolution network based on global dependency Transformer

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Abstract: At present, the image super-resolution network, based on deep learning, is mainly implemented by convolution. Compared with the traditional convolutional neural network, the main advantage of the Transformer in the image super-resolution task is its long-distance dependency modeling ability. However, most Transformer-based image super-resolution models cannot establish global dependencies when the number of parameters is small and the number of network layers is small, which limits the performance of the model. In order to establish global dependency in super-resolution network, an image super-resolution network based on the global dependency Transformer is proposed. Its main component is the residual square axial window block, and the internal axial window Transformer residual layer uses axial window and self-attention, which can make each pixel globally dependent on the entire feature map. In addition, the super-resolution image reconstruction module of most current image super-resolution models is composed of convolutions. In order to dynamically integrate the extracted feature information, Transformer and convolution are combined to jointly reconstruct super-resolution images. Experiments show that the image super-resolution network based on the global Transformer has an improvement of more than 0.3dB in the peak signal to noise ratio index on large-scale datasets compared with other advanced models in various super-resolution tasks.

Keywords: image super-resolution; Transformer; self-attention; global dependency; axial window

0 引言

单图像超分辨率(Single Image Super-Resolution, SISR)是计算机视觉和图像处理中的一个经典问题, 目标由低分辨率(Low Resolution, LR)图像生成对应的高分辨率图像(High Resolution, HR)。近年来, 基于卷积神经网络(Convolutional Neural Network, CNN)的图像超分辨率模型^[1-6]因其恢复图像

高频信息的能力而备受欢迎, 并在图像超分辨率领域占据主导地位。

近几年, 由于 Transformer^[7]在自然语言处理(Natural Language Processing, NLP)领域的广泛应用, Transformer 逐渐在计算机视觉领域被研究和使用的, 取得了非常显著的效果。相对于传统的卷积神经网络(CNN), Transformer 具有长距离依赖建模的优势, 在低级视觉任务, 尤其图像超分辨率领域具有非常广泛的应用前景。Transformer 的自注意力的计算量

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与参与计算的像素点数量成 2 次方关系。为了减少计算量, SwinIR^[8]基于 Swin Transformer^[9]把特征图划分为不重叠的方形窗口,在方形窗口内部进行自注意力计算,并使用移动窗口策略建立窗口间的联系,取得了超越基于 CNN 的模型性能。

尽管 SwinIR^[8]在图像超分辨率领域取得了很大的成功,但其存在一个固有的缺陷:在方形窗口内部进行自注意力仅仅会让模型有良好的局部信息处理能力。使用移动窗口策略,如图 1 所示,与使用卷积类似,随着网络层的堆叠,像素 A 建立依赖的区域会逐层扩大,在网络层数少、参数量小的网络中无法建立全局依赖,这会限制网络的性能。

针对上述问题,本文提出了基于全局依赖 Transformer 的图像超分辨率网络(Image Super-Resolution Network Based on Global Dependency Transformer, GDTSR)。根据 Ho 等^[10]提出的轴向窗口,本文提出了轴向窗口 Transformer 残差层(Axial Window Transformer Residual Layer, AWTRL),经过 2 个方向的轴向窗口内部的自注意力后,每个像素可以与整个特征图的所有像素点进行直接或间接地交互,与整个特征图建立起全局依赖。如图 2 所示,对于任意像素点 B,像素 A 可以通过两方向的轴向窗口内部自注意力(图中黑色窗口)与其建立依赖,像素 A 可以用类似的方式与整个特征图的所有像素建立依赖,即全局依赖。同时本文注意到,仅使用轴向窗口会使每个像素无法与临近像素直接进行自注意力,导致模型局部信息处理能力变差,因此本文兼顾使用与 SwinIR^[8]类似的方形窗口 Transformer 层(Square Window Transformer Layer, SWTL)。由 AWTRL 和 SWTL 共同组成的残差方形轴向窗口块(Residual Square-Axial Window Block, RSAWB)作为 GDTSR 的主要组成部分。

此外,目前单图像超分辨率网络的超分辨率重建模块大多由亚像素卷积^[11]或类似的结构组成。相比于与特征信息无关的卷积,Transformer 可以给不同的特征信息自适应地添加权重,使更有效的纹理信息发挥更大的作用,有利于重建清晰的超分辨率图像。因此本文把 Transformer 应用到超分辨率重建模块并设计了 Transformer 卷积上采样(Transformer and Convolution Upsample, TACUpSample)块。

如图 3 所示,本文提出的 GDTSR 与其他先进的参数量相近的模型相比,性能达到了最优。

本文的主要工作有:

1)提出了一个基于全局依赖 Transformer 的图像超分辨率网络(GDTSR)。该网络把轴向窗口与方形窗口相结合,让每个像素与整个特征图建立依赖,获得整个特征图信息的补充。

2)本文提出了一种通用的 Transformer 卷积上采样块(TACUpSample),把 Transformer 引入到超分辨率重建模块与卷积一起重建高分辨率图像。

3)实验表明,本文提出的 GDTSR 结合 TACUpSample 后,与其他先进的模型相比,在 3 个倍数的超分辨率实验上,模型性能在各个指标上都有很大提升。

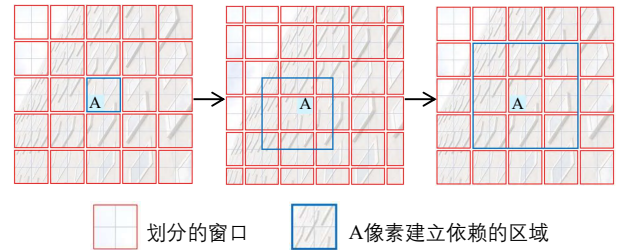


图 1 移动窗口策略

Fig.1 Strategy of shift window

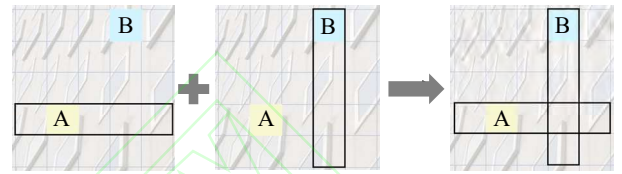


图 2 轴向窗口

Fig.2 Axial window

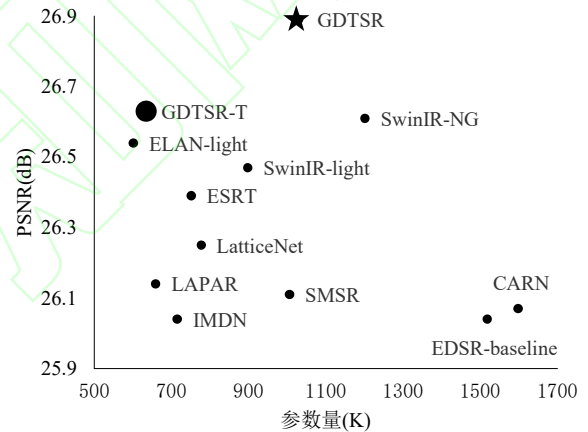


图 3 在 Urban100^[12]数据集上,各种超分辨率模型 4 倍 SR 的参数量和 PSNR 比较

Fig.3 PSNR and parameter comparison of different methods for $\times 4$ SR on dataset Urban100^[12]

1 相关工作

1.1 单图像超分辨率

单图像超分辨率是计算机视觉和图像处理中的一个经典问题。与传统的图像超分辨率方法^[13-14]相比,基于学习的方法,由于其优秀的性能而越来越受欢迎。这些方法大多从大规模的配对数据集中学习低分辨率图像到高分辨率图像之间的映射,自开创性工作 SRCNN^[1]出现后,一系列通过使用更复杂基于卷积神经网络架构的模型被提出,如残差块^[4]和稠密块^[6]等。部分研究使用了卷积神经网络内的注意力机制,如通道注意力^[5]和非局部注意力^[15]等。一些工作也把不同的框架应用在超分辨率领域上,如递归神经网络^[16]等。为了生成更符合人眼视觉的图片, Ledig 等^[17]和 Wang 等^[18]把生成

对抗网络(Generative Adversarial Network, GAN)^[19]应用到图像超分辨率领域。目前, 鉴于 Transformer 在高级视觉领域的成功应用, 越来越多的研究把 Transformer 应用在图像超分辨率网络中, 如 SwinIR^[8], EDT^[20]等, 取得了超过卷积神经网络的性能。

1.2 视觉 Transformer

近年来, Transformer^[7]在高级视觉任务领域取得了很大的成功, 包括图像分类^[9-10,21-23]、目标检测^[21-24]等。由于 Transformer 的自注意力计算量与参与计算的像素数量成 2 次方关系, 这么庞大的计算量无法使整张图片的所有像素点参与运算。为了降低计算量, DOSOVITSKIY 等^[21]提出 ViT, 把图像划分为 16×16 的窗口, 在窗口内部进行自注意力, 通过这种方式可以明显地减少其计算量。Chen 等^[25]把 ViT 用于图像超分辨率任务, 提出 IPT 模型, 通过超过 100M 的参数量, 取得了超越 CNN 的性能。虽然 IPT 在性能上取得了突破, 但是基于局部窗口划分, 只能取得很好的局部信息处理能力, 无法建立起窗口之间的依赖。Liu 等^[9]提出的 Swin Transformer 改善了上述问题, 在局部窗口的基础上使用了移动窗口策略, 通过这种方式可以在不增加计算量的基础上建立窗口间的交互。Ho 等^[10]把特征图按照两个轴的方向进行划分, 得到轴向窗口, 这种方式虽然可以建立全局依赖, 但是局部信息处理能力又会明显不足。Liang 等^[8]基于 Swin Transformer^[9]提出了 SwinIR, 在参数量明显小于 IPT 的情况下, 取得了超越 IPT 的性能。EDT^[20]把 SwinIR 中的方形窗口改进为矩形窗口, 并使用预训练进一步改进网络性能。ACT^[26]在 Transformer 的基础上添加卷积, 使二者关注图像的不同细节, 使模型获得了超越 SwinIR 的性能, 但需要更大的参数量作为代价。ESRT^[27]提出了卷积与 Transformer 的串行架构, 并使模型更加轻量化。此外还有使用图像多尺度特征的 NGswin^[28]和用于人脸超分辨率的 MSwinSR^[29]等相继被提出。虽然目前 Transformer 在图像超分辨率领域有广泛的应用, 但是如图 1 所示, 使用移动窗口策略只能使感受野逐层增加, 在参数量小, 层数少的网络中无法建立起全局依赖, 这会限制网络的性能。

2 网络设计

2.1 基于全局依赖 Transformer 的图像超分辨率网络

如图 4(a)所示, 基于全局依赖 Transformer 的超分辨率网络(GDTSR)由浅层特征提取模块、深层特征提取模块和超分辨率重建模块组成。

浅层特征提取模块: 把低分辨率图像 $I_{LR} \in \mathbb{R}^{H \times W \times 3}$ (H 和 W 为低分辨率图像高和宽包含的像素数量)作为输入。本文使用 3×3 的卷积 Conv(·) 来提取浅层特征, 过程为:

$$F_0 = \text{Conv}(I_{LR}) \quad (1)$$

其中 $F_0 \in \mathbb{R}^{H \times W \times C}$ 为图像的浅层特征, C 为特征图的通道数, 卷积操作具有良好的局部信息提取能力, 可以有效地提取图像的浅层特征。

深层特征提取模块: 图像经过浅层特征提取后, 从得到的浅层特征 F_0 中提取深层特征, 深层特征提取模块包含 K 个残差方形轴向窗口块(Residual Square-Axial Window Block, RSAWB)和一个 3×3 的卷积 Conv(·)。中间特征 $F_1, F_2, \dots, F_K$ 和最终输出的深层特征 F_D 都是逐块提取的, 过程为:

$$\begin{aligned} F_k &= f_{RSAWB}^k(F_{k-1}) \quad k=1, 2, \dots, K \\ F_D &= \text{Conv}(F_K) \end{aligned} \quad (2)$$

其中 $f_{RSAWB}^k(\cdot)$ 是第 k 个 RSAWB 函数。

超分辨率重建模块: 由于卷积核与作用的特征图无关, 这就导致卷积会平等地对待提取到的特征图中的不同信息, 根据 SwinIR^[8], Transformer 可以给不同的特征信息自适应地添加权重, 动态地使用特征图中的信息, 给更有效的纹理信息赋予更大的权重, 有利于提升重建超分辨率图像的能力。因此把 Transformer 引入到超分辨率图像重建模块, 设计了 Transformer 卷积上采样(Transformer and Convolution Upsample, TACUpSample)块作为超分辨率重建模块。TACUpSample 通过融合浅层特征 F_0 和深层特征 F_D 来重建超分辨率图像 I_{SR} , 过程为:

$$I_{SR} = f_{TAC}(F_D + F_0) \quad (3)$$

其中 $f_{TAC}(\cdot)$ 是 TACUpSample 函数, $I_{SR} \in \mathbb{R}^{sH \times sW \times 3}$, s 是图像超分辨率倍数。

2.2 轴向窗口 Transformer 残差层

SwinIR^[8]等目前大多数基于 Transformer 的图像超分辨率网络, 把特征图划分为局部窗口, 并且使用移动窗口策略建立窗口间的联系。如图 1 所示, 这种方式只能逐层扩大感受野, 在参数量小的模型中无法建立起全局依赖。针对这个问题, 本文设计了轴向窗口 Transformer 残差层(Axial Window Transformer Residual Layer, AWTRL)。

如图 4(d)所示, 为了使每个像素充分利用全局信息, 与整个特征图建立依赖, 根据[10], 在 AWTRL 中按照 H 或 W 轴方向, 把特征图划分为轴向窗口。如图 2 所示, 在两个方向的窗口内部进行自注意力计算, 可以使每个像素与整个特征图的所有像素点建立依赖, 获得整个特征图所有像素信息的补充, 建立全局依赖。以 H 轴窗口为例, 把输入的特征图 $H \times W \times C$ 划分为 $N \times (1 \times W) \times C$, 其中 N 为划分的窗口数量(同时也是特征图 H 轴的像素数量, $N = H$), 每个窗口包含 $1 \times W$ 个像素, 每个像素在其所在的 H 轴窗口内部进行自注意力。对于每个输入窗口 $X \in \mathbb{R}^{1 \times W \times C}$, query、key、value 矩阵 $Q, K, V \in \mathbb{R}^{1 \times W \times C}$ 的计算方式分别为:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V \quad (4)$$

其中 $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{C \times C}$ 为可学习参数。自注意力过程 $\text{SA}(\cdot)$ 为:

$$\text{SA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}(\mathbf{Q}\mathbf{K}^T)\mathbf{V} \quad (5)$$

其中, $\text{Softmax}(\cdot)$ 是 Softmax 函数。

此外, 根据 ACT^[26], Transformer 和卷积共同使用可以使二者关注特征图的不同细节。本文在 AWTRL 中添加了由卷积, 激活函数(GELU)和通道注意力(Channel Attention, CA)^[5]共同组成的残差块(Residual Block, RB)。多头自注意力(multi-head self-attention, MHSA)与残差块得到的共同特征经过多层感知机(Multi-Layer Perceptron, MLP)^[8]进一步整合。每个阶段都使用层归一化和残差。假设 AWTRL 的输入特征为 $\mathbf{X}_{in}$, 整个过程可以表示为:

$$\begin{aligned} \mathbf{X}_m &= \text{MHSA}(\text{LN}(\mathbf{X}_{in})) + \text{RB}(\text{LN}(\mathbf{X}_{in})) + \mathbf{X}_{in} \\ \mathbf{X}_{out} &= \text{MLP}(\text{LN}(\mathbf{X}_m)) + \mathbf{X}_m \end{aligned} \quad (6)$$

其中, $\text{RB}(\cdot)$ 是残差块函数, $\text{LN}(\cdot)$ 是层归一化函数, $\text{MLP}(\cdot)$ 是多层感知机函数。MHSA $(\cdot)$ 是多头自注意力函数, 其计算

过程是: 把输入特征的通道平分为 d 份, 即头数为 d , 并行执行 d 次 $\text{SA}(\cdot)$ 后, 拼接得到的 d 个特征作为输出特征 $\mathbf{X}_{out}$ 。

如图 2 所示, 对于特征图内任意像素点 A, 假设坐标为 (a, b) , 首先与所有坐标为 (h, b) ($h = 1, 2, \dots, H$) 的像素点进行自注意力, 然后所有坐标为 (h, b) 的像素点与对应的 (h, w) ($w = 1, 2, \dots, W$) 的所有像素点进行自注意力。此时已经与整张特征图的所有像素点建立了依赖, 则说明任意像素点都可以与整个特征图建立依赖关系, 即全局依赖。

把特征图划分为 H 轴窗口或 W 轴窗口, 并在窗口内部进行自注意力, 其自注意力的计算量为 $6C^2HW + 4CH^2W$ (以 H 轴窗口为例), 在输入图像的 H 与 W 近似相等的情况下, 其计算量是图像像素数量的近似 1.5 次方。相比于只使用方形窗口划分的 SwinIR^[8]等, 尽管计算量有所上升, 但是可以使每个像素与整个特征图建立全局依赖。经过实验可以得到, 本文的方法比 SwinIR^[8]等基于移动窗口策略的模型相比, 在性能上有的较大的提升(见表 1, 表 2 和表 3)。

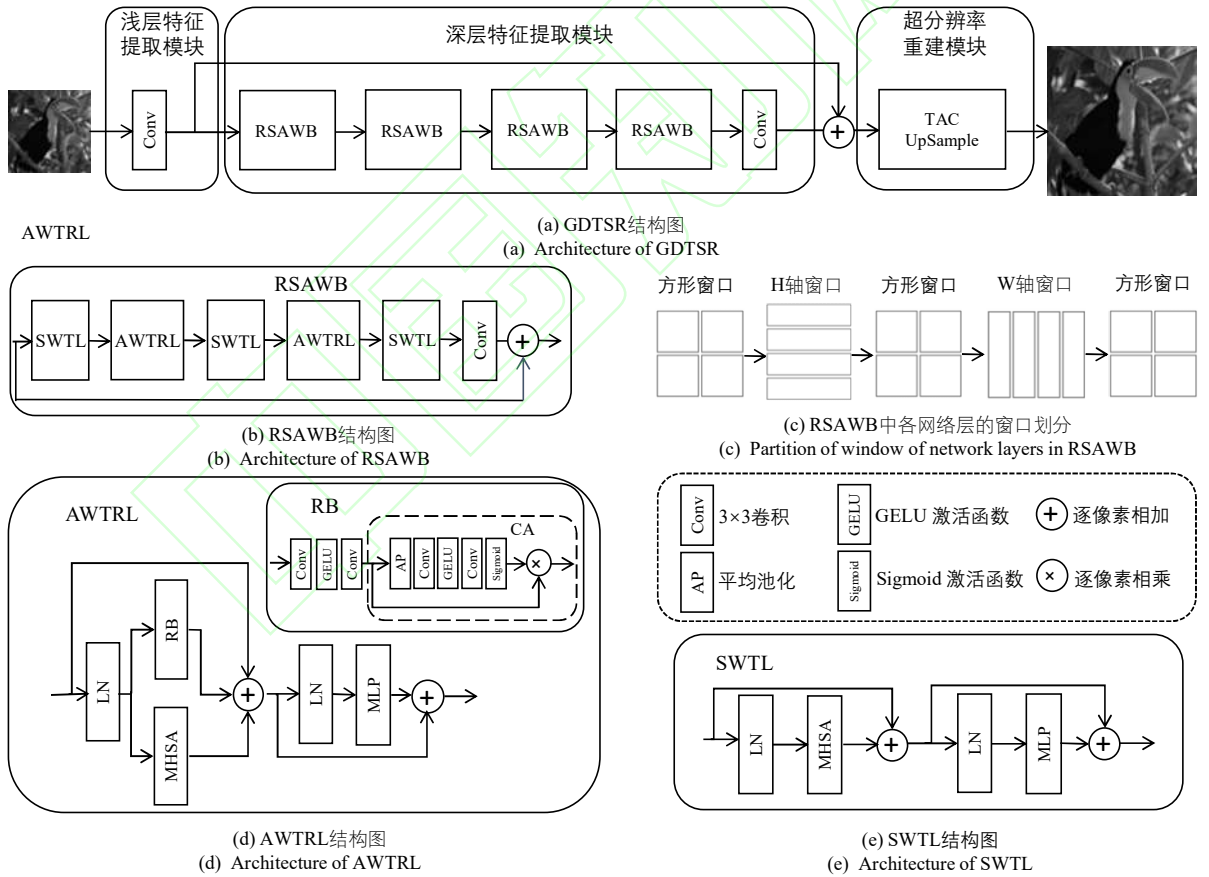


图 4 GDTSR、RSAWB、AWTRL 和 SWTL 的结构图
Fig.4 Architecture of GDTSR, RSAWB, AWTRL and SWTL

2.3 方形窗口 Transformer 层

在 AWTRL 中使用两个方向的轴向窗口可以使每个像素与整个特征图建立起全局依赖, 但是每个像素与大部分的邻

近像素没有直接进行自注意力, 这会导致模型的局部信息处理能力下降。为此, 本文设计了方形窗口 Transformer 层 (Square Window Transformer Layer, SWTL), 如图 4(e)所示。与 SwinIR^[8]的窗口划分方式相同, SWTL 把每个特征图划分为方形的局部窗口, 在窗口内部进行自注意力计算。整个过

程与 AWTRL 类似。并且为了轻量化, 在 SWTL 中不使用残差块, 以减少参数量。

2.4 残差方形轴向窗口块

由 AWTRL 和 SWTL 共同组成的残差方形轴向窗口块 (Residual Square-Axial Window Block, RSAWB), RSAWB 是本文提出模型的主要组成部分, 如图 4(b)所示。RSAWB 中的中间特征是被逐层提取的, 过程为:

$$\begin{aligned} F_{k,1} &= f_{S,1}^k(F_{k-1}) \\ F_{k,2} &= f_{A,H}^k(F_{k,1}) \\ F_{k,3} &= f_{S,2}^k(F_{k,2}) \\ F_{k,4} &= f_{A,W}^k(F_{k,3}) \\ F_{k,5} &= f_{S,3}^k(F_{k,4}) \\ F_k &= \text{Conv}(F_{k,5}) + F_{k-1} \end{aligned} \quad (7)$$

其中 $f_{S,j}^k(\cdot)$ 是第 k 个 RSAWB 中的第 j ($j=1,2,3$) 个 SWTL 函数, $f_{A,H}^k(\cdot)$ 和 $f_{A,W}^k(\cdot)$ 是第 k 个 RSAWB 中 H 轴向和 W 轴向的 AWTRL 函数。RSAWB 中网络层划分窗口方式如图 4(c)所示, RSAWB 使用了 AWTRL 和 SWTL 去提取深层特征, 其中 AWTRL 把特征图划分为轴向窗口, 相比于移动窗口策略, 经过两个方向的 AWTRL 后可以使每个像素与整个特征图建立依赖。SWTL 把特征图划分为方形窗口, 在方形窗口内部进行自注意力可以很好地提取特征的局部信息。同时使用 AWTRL 和 SWTL, 可以使模型保持良好的局部信息处理能力, 同时最大程度对每个像素进行全局信息的补充, 有利于提取到更有效的深层特征。

2.5 Transformer 卷积上采样块

目前大多数超分辨率模型的超分辨率重建模块都是使用卷积实现^[4,8,18,26], 如亚像素卷积^[11]等。由于卷积核与作用的特征图无关, 这就导致卷积会平等地对待提取到的特征图中的不同信息, 根据 SwinIR^[8], Transformer 在进行特征提取时, 可以给不同的特征信息自适应地添加权重, 相当于动态卷积。因此本文把 Transformer 引入到超分辨率重建模块, 与卷积共同一起进行超分辨率图像的重建, 如图 5 所示。整个 TACUpSample 作为超分辨率重建模块, 过程为:

$$\begin{aligned} F_M &= \text{Conv}(F_0 + F_D) + \text{MHSA}(\text{LN}(F_0 + F_D)) \\ I_{SR} &= \text{Conv}(\text{PixelShuffle}(F_M)) \end{aligned} \quad (8)$$

其中 PixelShuffle($\cdot$) 是 PixelShuffle 函数, 多头自注意力 MHSA($\cdot$) 计算过程与 AWTRL 中一致。

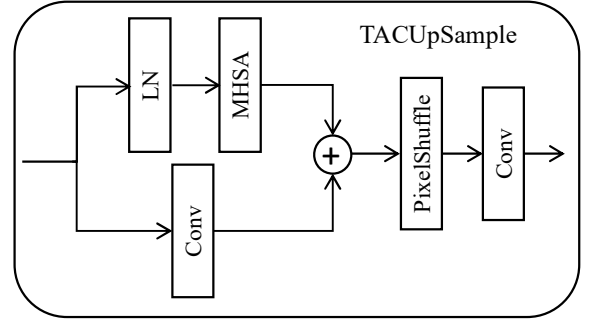


图 5 Transformer 卷积上采样块
Fig.5 Transformer and Convolution Upsample(TACUpSample)

2.6 损失函数

在单图像超分辨率任务中, 常用的损失函数为平均绝对误差^[4,5,8,20] (L_1 损失)和均方误差^[1] (L_2 损失)。相比于 L_2 损失, L_1 损失可以使生成的图像更加锐化, 有利于使超分辨率图像获得更清晰的纹理细节。与目前大多数超分辨率网络相同, 本文通过优化参数使生成的超分辨率图像与配对的高分辨率图像逐像素 L_1 距离最小, 为:

$$L_1 = \|I_{SR} - I_{HR}\|_1 \quad (9)$$

其中 I_{HR} 为配对的高分辨率图像。

3 实验与结果分析

3.1 实验设置

本文使用 DIV2K^[30]数据集中的 800 张训练图片作为训练集, 并对其随机进行多次 90° 旋转和水平翻转以实现数据增强。选取 DIV2K 的 851-860 图像作为验证集记为 DIV_val10。参考 SwinIR-light^[8]中 RSTB 的数量和 EDT-T^[20]中 Transformer Block 的数量均为 4, 本模型中 RSAWB 的数量同样设置为 4, 中间特征图的通道数 C 设置为 60, 多头自注意力的头数 d 设置为 6。为了与其他先进的超分辨率模型进行公平的比较, 本文设计了参数量为 600K(GDTSR-T)和 1M(GDTSR)的两个轻量级模型, 两个模型仅卷积的参数量不同。此外本文额外增加 DF2K(DIV2K+Flickr2K^[30])作为训练集的实验, 记为(GDTSR-DF), 进一步提升模型的性能。使用 $\beta_1 = 0.9$ 、 $\beta_2 = 0.999$ 、 $\varepsilon = 10^{-8}$ 的 ADAM^[31]优化器, 训练时每个批次处理 8 张 64×64 的训练图片, 初始学习率为 2×10^{-4} , 共训练 500000 次迭代, 在 250000、400000、450000、475000 次迭代后学习率降为之前的一半。在 Set5^[32]、Set14^[33]、B100^[34]、Urban100^[12]和 Manga109^[35]共 5 个标准数据集上测试模型性能。在 YCbCr 空间的亮度(Y)通道上, 使用峰值信噪比(Peak Signal to Noise Ratio, PSNR)^[36]和结构相似性(Structural Similarity Index, SSIM)^[36]来评估模型的性能。整个模型在 NVIDIA RTX A5000 上进行训练。

3.2 实验结果

3.2.1 客观指标

不同倍数($\times 2$, $\times 3$, $\times 4$)的超分辨率实验结果如表 1、表 2、表 3 所示, 最好和次好的结果分别用加粗和下划线表示。本文提出的 GDTSR 在 5 个标准测试集的测试结果中, 三个倍数的 PSNR 和 SSIM 结果, 与参数量相当的其他先进的超分辨率模型相比都达到了最优, 特别是在大尺寸图像的 Urban100 和 Manga109 数据集上模型性能的提升尤为明显。与 ELAN-light^[41]相比, 本文的 GDTSR-T 使用几乎相同的参数量, 在 Manga109 数据集上 PSNR 结果平均提升了 0.26dB。与参数量更大的 SwinIR-light^[8]相比, 本文的 GDTSR-T 在 Manga109 数据集上 PSNR 结果平均提升了 0.27dB。本文的 GDTSR-DF 与 EDT-T^[20]相比, 在 Urban100 数据集上的 PSNR 结果平均提升了 0.34dB, 在 Manga109 数据集上的 PSNR 结果平均提升了 0.29dB。在仅使用 DIV2K 作为训练集的情况下, 本文的 GDTSR 与 EDT-T^[20]相比仍有优势, 在 Manga109 数据集上的 PSNR 结果平均提升了 0.09dB, 与参数量相当的先进模型(如 SMSR^[43]、SwinIR-NG^[28]、Swin2SR-s^[45])相比, 各个数据集的 PSNR 结果均有较大提升。

表 1 各种 SISR 网络 2 倍 SR 的平均 PSNR 与 SSIM

Tab. 1 Average PSNR(dB) and SSIM of $\times 2$ SR for various SISR networks

训练集	模型	参数量	Set5		Set14		B100		Urban100		Manga109	
			PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM
DIV2K	IMDN ^[38]	694K	38.00	0.9605	33.63	0.9177	32.19	0.8996	32.17	0.9283	38.88	0.9774
DIV2K	LAPAR ^[39]	548K	38.01	0.9605	33.62	0.9183	32.19	0.8999	32.10	0.9283	38.67	0.9772
DIV2K	LatticeNet ^[40]	756K	38.15	0.9610	33.78	0.9193	32.25	0.9005	32.43	0.9302	--	--
DIV2K	ESRT ^[27]	677K	38.03	0.9600	33.75	0.9184	32.25	0.9005	32.58	0.9318	39.12	0.9774
DIV2K	SwinIR-light ^[8]	878K	38.14	0.9611	33.86	0.9206	32.31	0.9012	32.76	0.9340	39.12	0.9783
DIV2K	ELAN-light ^[41]	582K	38.17	0.9611	33.94	0.9207	32.30	0.9012	32.76	0.9340	39.11	0.9782
DIV2K	GDTSR-T	600K	38.17	0.9612	33.99	0.9203	32.31	0.9013	32.78	0.9342	39.27	0.9784
DIV2K	EDSR-baseline ^[4]	1370K	37.99	0.9604	33.57	0.9175	32.16	0.8994	31.98	0.9272	38.54	0.9769
DIV2K	CARN ^[42]	1592K	37.76	0.9590	33.52	0.9166	32.09	0.8978	31.92	0.9256	38.36	0.9765
DIV2K	SMSR ^[43]	985K	38.00	0.9601	33.64	0.9179	32.17	0.8990	32.19	0.9284	38.76	0.9771
DIV2K	HGSRNN ^[44]	2178K	37.80	0.9591	33.56	0.9175	32.12	0.8984	32.21	0.9292	--	--
DIV2K	SwinIR-NG ^[28]	1181K	38.17	0.9612	33.94	0.9205	32.31	0.9013	32.78	0.9340	39.20	0.9781
DIV2K	Swin2SR-s ^[45]	1000K	38.17	0.9613	33.95	0.9216	32.35	0.9024	32.85	0.9349	39.32	0.9784
DIV2K	GDTSR	1003K	<u>38.26</u>	<u>0.9615</u>	<u>34.13</u>	<u>0.9217</u>	32.35	0.9018	<u>33.15</u>	<u>0.9368</u>	<u>39.45</u>	0.9787
DF2K	EDT-T ^[24]	917K	38.23	<u>0.9615</u>	33.99	0.9209	<u>32.37</u>	<u>0.9021</u>	32.98	0.9362	<u>39.45</u>	0.9789
DF2K	GDTSR-DF	1003K	38.31	0.9616	34.28	0.9239	32.39	0.9023	33.34	0.9384	39.66	<u>0.9788</u>

表 2 各种 SISR 网络 3 倍 SR 的平均 PSNR 与 SSIM

Tab. 2 Average PSNR(dB) and SSIM of $\times 3$ SR for various SISR networks

训练集	模型	参数量	Set5		Set14		B100		Urban100		Manga109	
			PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM
DIV2K	IMDN ^[38]	703K	34.36	0.9270	30.32	0.8417	29.09	0.8046	28.17	0.8519	33.61	0.9445
DIV2K	LAPAR ^[39]	544K	34.36	0.9267	30.34	0.8421	29.11	0.8054	28.15	0.8523	33.51	0.9441
DIV2K	LatticeNet ^[40]	765K	34.53	0.9281	30.39	0.8424	29.15	0.8059	28.33	0.8538	--	--

3.2.2 主观效果

2 倍和 4 倍的超分辨率结果的主观效果如图 6 所示。以 Set14 数据集的 barbara 为例, 其他模型的结果都有模糊和纹理方向错误的问题, 本文模型恢复出的图像更加清晰, 纹理方向正确, 很接近于原 HR 图像。Urban100 数据集的 img092 也是类似。再以 Urban100 数据集的 img024 和 img059 为例, 其他模型恢复出的图像, 或过度模糊, 或严重失真, 本文的 GDTSR 可以恢复出更加清晰准确的纹理细节。

3.2.3 模型复杂度的比较

模型复杂度的比较如表 4 所示, 模型的计算量用 FLOPs 来表示, FLOPs 指模型在推理时乘法和加法运算的次数。由表 4 可以得到, 本文的模型与 SwinIR-light^[8]和 EDT-T^[20]相比, 计算量有所增加。这是因为本文的 AWTSR 中使用计算量更大的轴向窗口。本文弃用了移动窗口策略, 不需要在图像上进行移位操作, 同时也弃用了移动窗口策略使用的掩码矩阵。此外 GDTSR 使用了更少的网络层数, 这就导致虽然本文模型的计算量有所增加, 但是推理速度相当, 甚至推理时间有所减少。

DIV2K	ESRT ^[27]	770K	34.42	0.9268	30.43	0.8433	29.15	0.8063	28.46	0.8574	33.95	0.9455
DIV2K	SwinIR-light ^[8]	886K	34.62	0.9289	30.54	0.8463	29.20	0.8082	28.66	0.8624	33.98	0.9478
DIV2K	ELAN-light ^[41]	590K	34.61	0.9288	30.55	0.8463	29.21	0.8081	28.69	0.8624	34.00	0.9478
DIV2K	GDTSR-T	611K	34.62	0.9289	30.58	0.8463	29.23	0.8086	28.71	0.8629	34.35	0.9488
DIV2K	EDSR-baseline ^[4]	1555K	34.37	0.9270	30.28	0.8417	29.09	0.8052	28.15	0.8527	33.45	0.9439
DIV2K	CARN ^[42]	1592K	34.29	0.9255	30.29	0.8407	29.06	0.8034	28.06	0.8493	33.43	0.9427
DIV2K	SMSR ^[46]	993K	34.40	0.9270	30.33	0.8412	29.10	0.8050	28.25	0.8536	33.68	0.9445
DIV2K	HGSRNN ^[44]	2363K	34.35	0.9260	33.32	0.8413	29.09	0.8042	28.29	0.8546	--	--
DIV2K	SwinIR-NG ^[28]	1190K	34.64	0.9293	30.58	0.8471	29.24	0.8090	28.75	0.8639	34.22	0.9488
DIV2K	GDTSR	1014K	<u>34.78</u>	<u>0.9300</u>	<u>30.67</u>	<u>0.8477</u>	<u>29.29</u>	<u>0.8102</u>	<u>29.07</u>	<u>0.8691</u>	<u>34.58</u>	<u>0.9504</u>
DF2K	EDT-T ^[24]	919K	34.73	0.9299	30.66	<u>0.8481</u>	<u>29.29</u>	<u>0.8103</u>	28.89	0.8674	34.44	0.9498
DF2K	GDTSR	1014K	34.83	0.9304	30.72	0.8487	29.33	0.8109	29.22	0.8717	34.80	0.9513

表 3 各种 SISR 网络 4 倍 SR 的平均 PSNR 与 SSIM
Tab. 3 Average PSNR(dB) and SSIM of $\times 4$ SR for various SISR networks

训练集	模型	参数量	Set5		Set14		B100		Urban10		Manga109	
			PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM	PSNR/dB	SSIM
DIV2K	IMDN ^[38]	715K	32.21	0.8948	28.58	0.7811	27.56	0.7353	26.04/	0.7838	30.45	0.9075
DIV2K	LAPAR ^[39]	659K	32.15	0.8944	28.61	0.7818	27.61	0.7366	26.14	0.7871	30.42/	0.9074
DIV2K	LatticeNet ^[40]	777K	32.30	0.8962	28.68	0.7830	27.62	0.7367	26.25	0.7873	--	--
DIV2K	ESRT ^[27]	751K	32.19	0.8947	28.69	0.7833	27.69	0.7379	36.39	0.7962	30.75	0.9100
DIV2K	SwinIR-light ^[8]	897K	32.44	0.8976	28.77	0.7858	27.69	0.7406	26.47	0.7980	30.92	0.9151
DIV2K	ELAN-light ^[41]	601K	32.43	0.8975	28.78	0.7858	27.69	0.7406	26.54	0.7982	30.92	0.9150
DIV2K	GDTSR-T	627K	32.40	0.8978	28.84	0.7869	27.72	0.7414	26.64	0.8012	31.21	0.9163
DIV2K	EDSR-baseline ^[4]	1518K	32.09	0.8938	28.58	0.7813	27.57	0.7357	26.04	0.7849	30.35	0.9067
DIV2K	CARN ^[42]	1592K	32.13	0.8937	28.60	0.7806	27.58	0.7349	26.07	0.7837	30.42	0.9070
DIV2K	SMSR ^[46]	1006K	32.12	0.8932	28.55	0.7808	27.55	0.7351	26.11	0.7868	30.54	0.9085
DIV2K	HGSRNN ^[44]	2321K	32.13	0.8940	28.62	0.7820	27.60	0.7363	26.27	0.7908	--	--
DIV2K	SwinIR-NG ^[28]	1201K	32.44	0.8980	28.83	0.7870	27.73	0.7418	26.61	0.8010	31.09	0.9161
DIV2K	GDTSR	1030K	<u>32.65</u>	<u>0.9001</u>	<u>28.91</u>	<u>0.7885</u>	<u>27.78</u>	<u>0.7435</u>	<u>26.89</u>	<u>0.8084</u>	<u>31.47</u>	<u>0.9191</u>
DF2K	EDT-T ^[24]	922K	32.53	0.8991	28.88	0.7882	27.76	0.7433	26.71	0.8051	31.35	0.9180
DF2K	GDTSR	1030K	32.71	0.9007	28.97	0.7898	27.82	0.7444	27.04	0.8120	31.64	0.9202

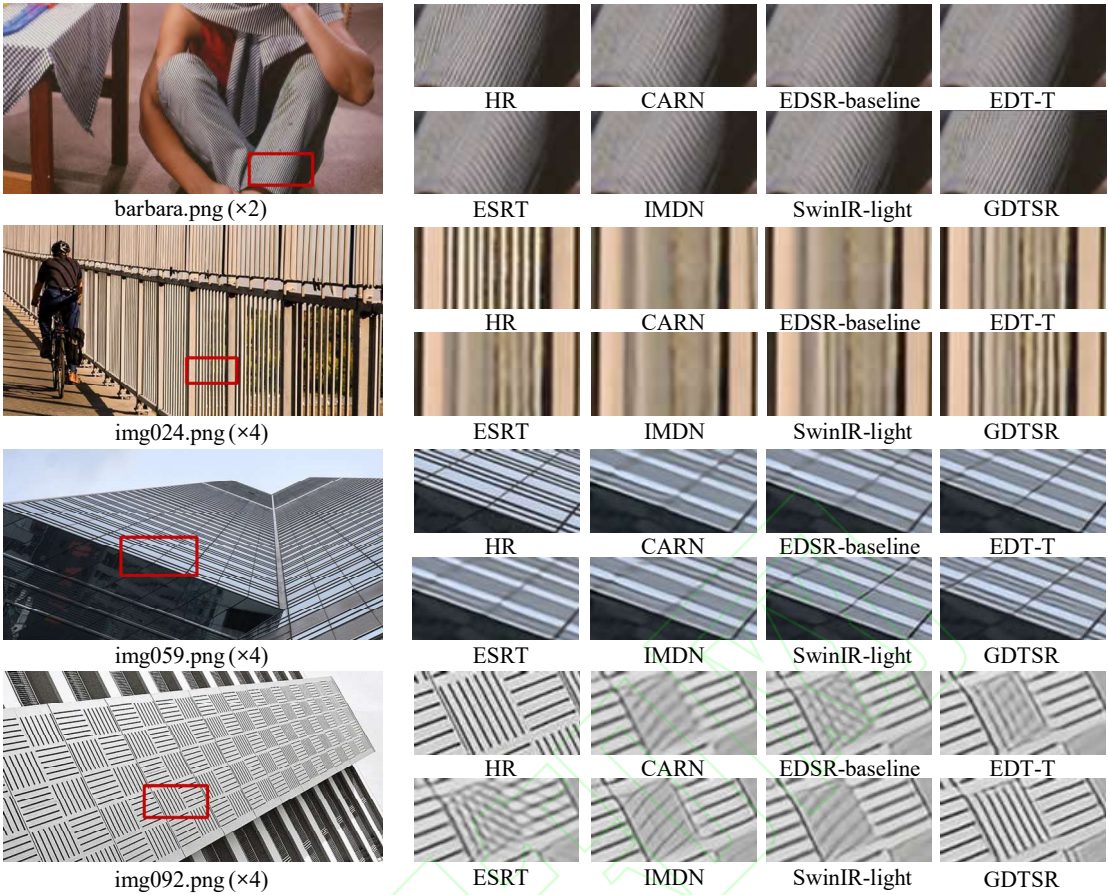


图 6 2 倍和 4 倍 SR 的视觉效果比较
Fig.6 Visual qualitative comparison of $\times 2$ and $\times 4$ SR

表 4 在 4 倍 SR 的 Urban100 数据集上, 3 种模型在推理时间、FLOPs 和 PSNR 的比较
Tab. 4 On dataset Urban100 of $\times 4$ SR, inference time, FLOPs and PSNR comparison in 3 models

模型	推理时间/s	GFLOPs	PSNR/dB
SwinIR-light ^[8]	118.25	49.6	26.47
EDT-T ^[20]	128.48	54.9	26.71
GDTSR	110.75	71.9	26.89

3.3 消融实验

3.3.1 轴向窗口的有效性

本文的 AWTRL 包含 H 或 W 轴的条形窗口, 为验证两个方向的轴向窗口提供的全局信息有利于超分辨率性能的提高。本文设计把 AWTRL 中的条形窗口替换为与 SWTL 中相同的方形局部窗口, 与本文模型进行比较, 结果如表 5 示。可以看到在整个模型中仅仅使用方形局部窗口, 虽然有更好的局部信息处理能力, 但是每个像素点无法取得较远位置信息的补充, 缺少全局依赖, 模型的性能也有较明显的下降。

表 5 AWTRL 中不同的窗口, 在验证集 DIV_val10 上, 4 倍 SR 的 PSNR 结果

Tab. 5 On validation dataset DIV_val10, PSNR results of $\times 4$ SR

with different window in AWTRL	
AWTRL 的窗口类型	PSNR/dB
方形窗口	29.72
轴向窗口	29.89

3.3.2 残差块的有效性

本文的 AWTRL 额外添加了由卷积、激活函数和通道注意力^[5]组成的残差块。为验证其有效性, 设计了在 AWTRL 中不使用残差块的模型, 记为 GDTSR_w/o_RB, 与本文模型进行对比, 实验结果见表 6。可以看到使用了残差块虽然参数量会有所增加, 但是模型的性能会有显著提高。根据 ACT^[26], 说明在参数量小的超分辨率网络中, 共同使用卷积与 Transformer, 同样可以使二者关注图片的不同细节, 有利于提取到更准确有效的高频信息。

表 6 在验证集 DIV_val10 上, ATRL 包含 RB 与不包含 RB, 4 倍 SR 的参数量和 PSNR 结果

Tab. 6 On validation dataset DIV_val10, parameter quantification and PSNR results of $\times 4$ SR with or without RB in AWTRL

模型	参数量/ 10^3	PSNR/dB
GDTSR_w/o_RB	773	29.79
GDTSR	1030	29.89

3.3.3 轴向窗口宽度的讨论

本文使用的轴向窗口为或，窗口的宽度为 1。为了讨论轴向窗口宽度对于模型的影响，本文在 AWTRL 中分别设置了宽度分别为 2、4 和 8 的轴向窗口，在性能、推理时间和占用显存上与本文模型进行比较，结果如表 7 所示。通过对比可以发现，轴向窗口宽度的增加，模型的推理时间和占用显存会有明显的提高，但模型的性能只有很小的提升。相比于使用宽度为 1 的轴向窗口，宽度设置为 8，其推理时间增加了近 60%，同时占用显存增加了近 2 倍。但 PSNR 仅仅增长了 0.02dB。很显然轴向窗口越大，其推理速度越慢，占用显存也会随之越大。本文认为使用任意宽度的轴向窗口都能起到建立全局依赖的效果。使用更大宽度的轴向窗口，会在窗口内部包含更多的局部信息，但 SWTL 中已经使用了方形的局部窗口，有较强的局部信息处理能力，在 AWTRL 中添加更多的局部信息不能使模型性能有较为明显的增加。本文把轴向窗口宽度设置为 1，可以在性能、推理速度和占用显存上有很好地权衡。

表 7 在验证集 DIV_val10 上，ATWSR 中使用不同宽度的轴向窗口，4 倍 SR 的占用显存、推理时间和 PSNR 结果
Tab. 7 On validation dataset DIV_val10, memory, inference time and PSNR results of $\times 4$ SR with different width of axial window in ATWSR

窗口宽度	占用显存/MB	推理时间/s	PSNR/dB
1	3898	50.46	29.89
2	5427	56.73	29.89
4	7982	62.94	29.90
8	9864	81.87	29.91

3.3.4 Transmre 卷积上采样块的有效性

为证明本文提出的 TACUpSample 比单独使用卷积可以取得更好的超分辨率重建效果，本文对于超分辨率重建模块进行了消融实验，实验设置了仅 TACUpSample 中的卷积和仅 TACUpSample 的 Transformer 作为超分辨率重建模块，与与本文模型进行比较，两个模型分别记为 GDTSR_0 和 GDTSR_1，相比较的三个模型参数量几乎相同，实验结果如表 8 所示。从实验结果可以看到，在仅使用 Transformer 时相比于其他模型，性能有较明显的下降，说明在超分辨率重建模块中使用卷积是必不可少的。用本文提出的 TACUpSample，比单独使用卷积，参数量只增加了 0.3%，PSNR 结果有 0.04dB 的提升，说明 Transformer 可以与卷积一起整合浅层和深层特征，相比于与只使用卷积可以更好地重建超分辨率图像。

此外，为了验证 TACUpSample 的通用性，本文把 EDSR-baseline^[4]和 SwinIR-light^[8]的超分辨率重建模块替换为 TACUpSample，实验结果如表 9 所示。可以看到，其他经典的轻量级超分辨率模型在使用 TACUpSample 后，可以提升模型性能。

表 8 在验证集 DIV_val10 上，GDTSR 中不同的超分辨率重建模块，4 倍 SR 的 PSNR 结果
Tab. 8 On validation dataset DIV_val10, PSNR results of $\times 4$ SR

with different super-resolution reconstruction in GDTSR

模型	PSNR/dB	参数量/ 10^3
GDTSR_0	29.85	1030
GDTSR_1	29.66	1024
GDTSR	29.89	1027

表 9 不同的模型中使用 TACUpSample，在验证集 DIV_val10 上，4 倍 SR 的 PSNR 结果
Tab. 9 On validation dataset DIV_val10, PSNR results of $\times 4$ SR with different model using TACUpSample

模型	TACUpSample	PSNR/dB
EDSR-baseline	-	29.61
	$\sqrt{}$	29.67($\uparrow 0.06$)
SwinIR-light	-	29.80
	$\sqrt{}$	29.83($\uparrow 0.03$)

3.3.5 训练数据集对网络性能的影响

为了说明在训练集中添加 Flickr2K 数据集，对于网络性能的影响，本文把仅使用 DIV2K 作为训练集的 GDTSR 和使用 DF2K(DIV2K+Flickr2K)的 GDTSR-DF 分别在验证集 DIV_val10 上进行了实验，结果如表 10 所示。可以看到，在训练集中添加了 Flickr2K 数据集后，对网络的性能会有很大提升，在验证集上有 0.29dB 的提升。在 5 个通用测试集上，如表 1、表 2、表 3 所示，网络性能均有很大程度的提高。

表 10 GDTSR 和 GDTSR-DF，在验证集 DIV_val10 上，4 倍 SR 的 PSNR 结果
Tab. 10 On validation dataset DIV_val10, PSNR results of $\times 4$ SR with GDTSR and GDTSR-DF

模型	PSNR/dB
GDTSR	29.89
GDTSR-DF	30.18

4 结语

本文提出了基于全局依赖 Transformer 的图像超分辨率网络 GDTSR，GDTSR 使用两个方向的轴向窗口建立全局依赖，比目前基于移动窗口策略的模型^[8,20]，能更好地利用整个特征图的所有信息。GDTSR 兼顾使用方形窗口来保持良好的局部信息处理能力，两者结合可以提取更有效的高频信息。此外本文提出了通用的卷积与 Transformer 结合的超分辨率图像重建模块，比起单独使用卷积可以更好地整合提取到的特征，提升模型的超分辨率重建能力。实验表明，与其他先进的超分辨率模型相比，GDTSR 在各个尺度的测试集上的性能达到了最优，特别是在大尺寸纹理细节丰富的数据集上展示出了较大的优势。

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基于像素对比学习的图像超分辨率算法

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摘要 目前, 深度卷积神经网络 (Convolutional neural network, CNN) 已主导了单图像超分辨率 (Single image super-resolution, SISR) 技术的研究, 并取得了很大进展. 但是, SISR 仍是一个开放性问题, 重建的超分辨率 (Super-resolution, SR) 图像往往会出现模糊、纹理细节丢失和失真等问题. 提出一个新的逐像素对比损失, 在一个局部区域中, 使 SR 图像的像素尽可能靠近对应的原高分辨率 (High-resolution, HR) 图像的像素, 并远离局部区域中的其他像素, 可改进 SR 图像的保真度和视觉质量. 提出一个组合对比损失的渐进残差特征融合网络 (Progressive residual feature fusion network, PRFFN). 主要贡献有: 1) 提出一个通用的基于对比学习的逐像素损失函数, 能够改进 SR 图像的保真度和视觉质量; 2) 提出一个轻量的多尺度残差通道注意力块 (Multi-scale residual channel attention block, MRCAB), 可以更好地提取和利用多尺度特征信息; 3) 提出一个空间注意力融合块 (Spatial attention fuse block, SAFB), 可以更好地利用邻近空间特征的相关性. 实验结果表明, PRFFN 显著优于其他代表性方法.

关键词 图像超分辨率, 卷积神经网络, 对比学习, 注意力机制

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Pixel-wise Contrastive Learning for Single Image Super-resolution

ZHOU Deng-Wen¹ LIU Zi-Han¹ LIU Yu-Kai¹

Abstract Deep convolutional neural network (CNN) has achieved great success in single image super-resolution (SISR). However, SISR is still an open issue, and reconstructed super-resolution (SR) images often suffer from blurring, loss of texture details and distortion. In this paper, a new pixel-wise contrastive loss is proposed to improve the fidelity and visual quality of SR images by making the pixels of SR images as close as possible to the corresponding pixels of the original high-resolution (HR) images and away from the other pixels in the local region. We also propose a progressive residual feature fusion network (PRFFN) with combined contrastive loss, and the main contributions include: 1) A general pixel-wise loss function based on contrastive learning is proposed, which can improve the fidelity and visual quality of SR images; 2) A lightweight multi-scale residual channel attention block (MRCAB) is proposed, which can better extract and utilize multi-scale feature information; 3) A spatial attention fusion block (SAFB) is proposed, which can better utilize the correlation of neighboring spatial features. The experimental results demonstrate that PRFFN significantly outperforms other representative methods.

Key words Image super-resolution, convolutional neural network (CNN), contrastive learning, attention mechanism

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单图像超分辨率 (Single image super-resolution, SISR)^[1] 是计算机视觉中一个基本任务, 旨在从低分辨率 (Low-resolution, LR) 图像, 恢复出对应的高分辨率 (High-resolution, HR) 图像, 在诸如遥感成像^[2]、视频监控^[3] 和医学成像^[4] 中, 应用广泛. SISR 是一个病态的逆问题, 因为许多 HR 图像可退化为相同的 LR 图像, 需要提供图像的先验知识,

约束超分辨率 (Super-resolution, SR) 图像的解空间. SISR 仍然是计算机视觉中开放性的研究问题, 重建的 SR 图像往往会出现模糊、纹理细节丢失和失真等问题.

早期的 SISR 是基于插值的方法, 如双线性插值和双三次 (Bicubic) 插值. 基于插值的方法仍被广泛使用, 具有很低的计算复杂度, 但不能恢复 LR 图像中丢失的图像细节. 基于实例学习的方法^[5–7] 旨在通过训练图像, 学习 LR 图像和 HR 图像之间的映射关系, 改进了基于插值的方法. 但是, 基于实例学习的方法往往优化困难, 并具有较高的推理复杂度. 目前, 深度卷积神经网络 (Convolutional neural

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networks, CNN) 技术^[8-13] 直接端到端地学习 LR 和 HR 图像之间的映射关系, 显著提高了 SISR 性能, 并主导了目前 SISR 技术的研究. 基于 CNN 的 SISR 方法研究主要是探索新的 SR 网络架构, 损失函数广泛使用均方误差 (Mean squared error, MSE)^[12] 和平均绝对误差 (Mean absolute error, MAE)^[8], 但这些传统的逐像素损失生成的 SR 图像是潜在 SR 输出图像的平均^[14], 导致输出的图像被过度平滑. 为了解决这个问题, Johnson 等^[15] 提出感知损失. 感知损失不是在图像空间度量逐像素的损失, 而是在预训练的 VGG (Oxford visual geometry group) 网络^[16] 特征空间度量逐像素损失. 感知损失能改进 SR 图像的感知质量, 但降低了 SR 图像保真度. 重要的是, 感知损失也不能阻止 SR 图像的模糊. Wang 等^[17] 提出一个对比自蒸馏 (Contrastive self-distillation, CSD) 网络, 引入一个基于对比学习的损失函数. CSD 损失与感知损失类似, 也在预训练 VGG 网络的特征空间逐像素比较损失. 以教师子网络输出的 SR 图像作为正样本, 学生子网络输出的 SR 图像作为锚, 从同一个批次中采样 K 个图像 (除锚外), 通过双三次上采样到与输出 SR 图像相同的分辨率作为负样本. CSD 损失使锚更靠近正样本, 并远离负样本, 进一步改进了学生子网络输出的 SR 图像的视觉质量. 与感知损失相比, CSD 损失除限制了学生子网络输出的上界 (正样本) 外, 也限制了下界 (负样本), 以减小解空间. 但 CSD 损失也有与感知损失类似的保真度低问题. 另外, CSD 损失使用双三次上采样图像作为负样本, 是一个较弱的下界. 受 CSD 启发, 提出一个新的基于对比学习的逐像素损失函数 L_{cntr} . L_{cntr} 作用在图像空间, 原 HR 图像与输出的 SR 图像对应的像素分别作为正样本和锚, HR 图像上正样本邻近的像素作为负样本. 类似地, L_{cntr} 也让锚更靠近正样本, 远离负样本. 常用损失及其组合的峰值信噪比 (Peak signal to noise ratio, PSNR)^[18]、结构相似性 (Structural similarity, SSIM)^[18] 和视觉效果见图 1, 其中 L_1 为 MAE 损失, L_{perc} 为感知损失^[15], L_{tex} 为纹理损失^[19], L_{CSD} 为 CSD 损失^[17], L_{adv} 为对抗损失^[20]. 原 L_{CSD} 正样本是教师网络输出的 SR 图像, 本文替换为 HR 图像. 计算 L_{tex} 的图像块大小为 48×48 像素. 可以看出, 本文的逐像素损失函数 L_{cntr} 可与其他损失组合使用, 显著改进 SR 图像保真度和视觉质量. 另外, 本文提出一个新的网络架构, 称为渐进残差特征融合网络 (Progressive residual feature fusion network, PRFFN). PRFFN 应用扩张卷积^[21], 扩展极深残差通道注意力网络 (Very deep residual channel attention network, RCAN)^[11] 的基本构件——

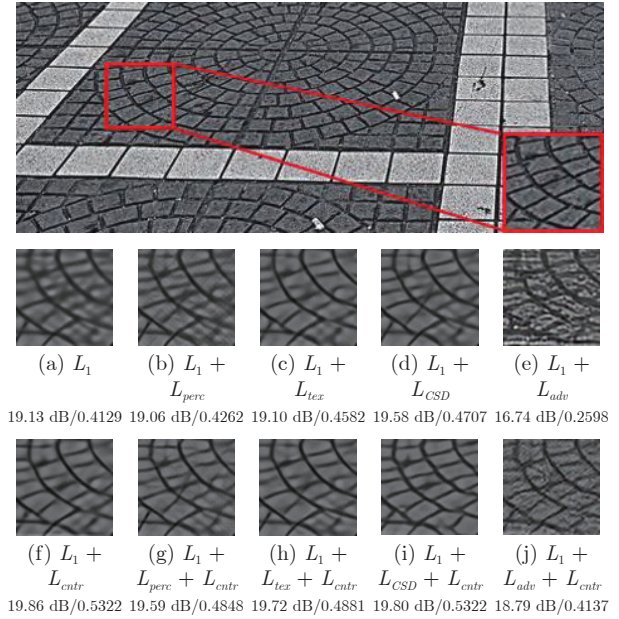


图 1 不同损失及其组合的 PSNR/SSIM 和视觉效果

Fig. 1 PSNR/SSIM and visual effects for different losses and their combinations

残差通道注意力块 (Residual channel attention block, RCAB)^[11], 不增加参数并融合多尺度特征, 称为多尺度残差通道注意力块 (Multi-scale residual channel attention block, MRCAB). PRFFN 以 MRCAB 为基本构件, 并运用空间注意力机制^[22], 渐进地融合 MRCAB 的输出特征. 本文的 L_{cntr} 损失与 PRFFN 网络架构组合, 能够获得先进的 SR 性能, 一些代表性方法的 PSNR 和参数量见图 2. L_{cntr} 是通用的, 可与其他 SR 网络架构协作使用.

本文的主要贡献有: 1) 提出一个通用的基于对比学习的逐像素损失函数 L_{cntr} , 能够显著改进 SR

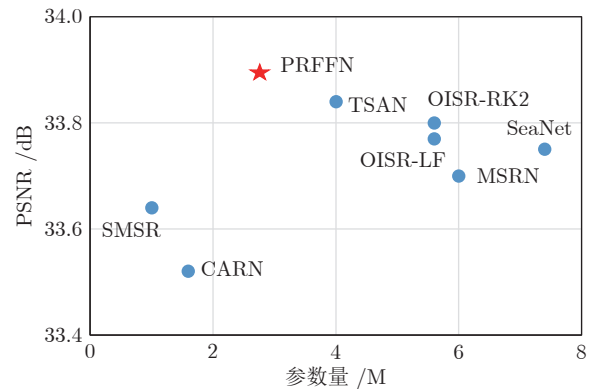


图 2 在 Set14 数据集上, 不同 SISR 方法 2 倍 SR 结果的平均 PSNR 值和参数量

Fig. 2 Average PSNRs and parameter counts for 2 times SR models for each state-of-the-art SISR method on the Set14 dataset

图像的视觉质量; 2) 提出一个新的 SR 网络架构 PRFFN, 主要组件是 MRCAB 和空间注意力融合块 (Spatial attention fuse block, SAFB), MRCAB 可以更好地提取和利用多尺度特征信息, 而 SAFB 可以更好地利用邻近特征的相关性; 3) 实验结果表明, PRFFN 组合 L_{cntr} 取得了有竞争力的 SR 性能.

1 相关工作

1.1 图像超分辨率的网络架构

2015 年, Dong 等^[8] 提出第 1 个基于 CNN 的 SR 方法——超分辨率卷积神经网络 (Super-resolution convolutional neural network, SRCNN) 后, 深度卷积神经网络技术已主导了当前 SR 技术研究. SRCNN 只有 3 个卷积层, Kim 等^[9] 提出极深超分辨率网络 (Very deep super-resolution network, VDSR), 通过引入残差学习, 加深了网络 (有 20 个卷积层), 改进了性能. Lim 等^[10] 提出增强的深度超分辨率网络 (Enhanced deep super-resolution network, EDSR), 通过堆积残差块, 进一步加深了网络 (有 65 个卷积层), 改进了性能. Zhang 等^[11] 提出 RCAN 方法, 在残差块中引入通道注意力 (Channel attention, CA) 机制, 有超过 400 个卷积层, 显著改进了 SISR 方法性能. Tong 等^[12] 提出超分辨率稠密连接网络, 使用了稠密连接. Zhang 等^[23] 提出残差稠密网络, 同时使用残差稠密连接. 稠密连接和残差稠密连接比残差连接可以更好地利用深度卷积层的分层特征. 同样, 注意力机制在 SR 网络中也得到了广泛关注, 通道注意力机制^[11] 考虑不同通道特征之间的依赖, 显著提高了模型的表示能力和 SR 性能. Zhang 等^[13] 提出一种非局部注意力块, 以建模长距离像素之间的依赖, 可以很好地捕捉空间注意力, 进一步增强特征的表示能力. Niu 等^[24] 提出整体注意力网络, 进一步组合了层注意力、通道和空间注意力, 以建模层、通道和空间位置之间的整体依赖. Li 等^[25] 提出多尺度残差网络 (Multi-scale residual network, MSRN), 采用局部多路径学习, 同时提取多个尺度特征, 改进了特征的表示能力. 视觉 Transformer 能够建模全局像素之间的依赖, 在图像分类、物体检测和分割等高级视觉任务中, 取得了巨大的成功. Liang 等^[26] 将 Swin Transformer^[27] 引入图像恢复, 取得了最先进的性能. 这些代表性的 SISR 方法虽然有好的 SR 性能, 但模型参数量较大, 需要较大的内存和较强的计算能力, 限制了它们在资源受限设备上的应用. 目前, 网络架构设计的趋势是设计更轻量的网络模型, 找到网络模型复杂度和性能之间最优的平衡^[28]. 本文模型

的基本构件设计受 Zhang 等^[11] 和 Li 等^[25] 启发, 相较于 RCAN 的基本构件 RCAB, 通过引入扩张卷积, 可以提取多尺度特征^[25] 且不增加参数量; 同时, 设计一个空间注意力融合块, 可以有效地融合这些多尺度的特征.

1.2 图像超分辨率的损失函数

目前, 基于 CNN 的 SR 技术研究大多聚焦于网络架构设计, 但合适的损失函数对 SR 模型的性能也至关重要. 最广泛使用的损失函数是逐像素的 L_1 损失 (即平均绝对误差) 和 L_2 损失 (即均方误差). 这些逐像素的误差度量返回许多可能解的平均值^[14], 会导致 SR 图像出现模糊、过度平滑和不自然的外观等问题, 尤其是在信息丰富的区域. 为了改进 SR 图像的感知质量, Johnson 等^[15] 提出感知损失, 使用预训练的图像分类网络, 在特征空间度量高级感知和语义差异. 与 L_1 等像素损失不同, 感知损失鼓励输出的 SR 图像与目标图像具有相似特征表示, 而不是迫使其像素匹配. 考虑到 SR 图像应与目标图像具有相同的纹理, Sajjadi 等^[19] 提出一种基于块的纹理损失, 图像纹理视为不同特征通道之间的相关性, 其定义基于预训练分类网络特征的 Gram 矩阵. 纹理损失迫使 SR 图像应与目标图像之间局部纹理相似. Ledig 等^[20] 提出对抗性损失, 鼓励 SR 图像逼近自然图像流形. 这些损失函数改进 SR 图像的视觉质量, 但是保真度低, 仍不能阻止图像模糊.

1.3 对比学习

对比学习^[29-30] 旨在学习一个嵌入的空间, 使得相似的样本彼此靠近, 不同的样本彼此远离. 对比学习可用于监督的学习环境, 也可用于无监督的学习环境, 在图像分类和图像聚类等高级视觉任务中, 应用非常成功, 但很少用于低级视觉任务. 高级视觉任务的对比学习技术可能不适用于低级视觉任务, 因为前者更需要全局视觉表示, 而后者更需要丰富的纹理细节^[31]. 对比学习需要考虑度量相似性的隐空间和正/负样本的选取两个重要因素. 在低级视觉任务中, 使用的隐空间大都是预训练网络 (例如 VGG) 的特征空间. 采用简单方法选取正/负样本, 例如以输入的低质量图像作为负样本, 以输入的高质量图像作为正样本. Wu 等^[32] 提出图像去雾方法, 使用 VGG 特征空间真实图像作为正样本, 退化的雾图像作为负样本, 估计的图像作为锚. 在 VGG 特征空间中, 让锚靠近正样本, 远离负样本. Wang 等^[17] 提出一个对比自蒸馏 SR 网络, 同时使用 VGG 特征空间, 其中教师子网络输出的 SR 图像作为正样本, 学生子网络输出的 SR 图像作为锚,

从同一个批次中采样 K 个图像 (除锚外), 通过双三次插值上采样到与输出 SR 图像相同的分辨率, 作为负样本. 在盲 SR 中, Wang 等^[33] 使用一个编码器子网络, 学习不同退化的抽象表示. 假定相同图像中的块退化相同, 不同图像中的块退化不同, 以分别选择为正/负样本. Wu 等^[31] 提出一个基于对比学习的 SR 框架, 以判别器子网络的特征空间作为隐空间. 通过对真实图像施加一些轻微的模糊, 生成多个难的负样本并简单锐化真实图像, 生成多个信息丰富的正样本. Wu 等^[31] 的隐空间对退化更敏感, 其负样本试图促使 SR 图像远离平滑的结果, 而其正样本试图迫使 SR 图像吸收更多的细节信息. 本文受到 CSD 的启发, CSD 虽能改进视觉结果, 但保真度低, 正/负样本限定的上/下界距离大, 即上/下界较弱. 本文直接使用图像空间训练 HR 图像和输出的 SR 图像对应的像素, 分别作为正样本和锚; HR 图像上正样本邻近的像素, 作为负样本.

2 本文方法

本文的渐进残差特征融合网络架构如图 3 中模块 1) 所示, 主要包括浅层特征提取块 (Shallow feature extraction block, SFEB)、特征映射块 (Feature mapping block, FMB) 和上采样块三部分. 其中, SFEB 仅包含一个 3×3 的卷积层, 提取浅层特征信息; FMB 包含 N (实验中, $N = 10$) 个渐进特征融合组 (Progressive feature fuse group, PFFG);

上采样块使用亚像素卷积^[34], 使用 L_1 损失和本文的像素级对比损失 L_{ctr} . 假定输入的 LR 图像为 I_{LR} , 首先, 经过 SFEB 提取浅层特征:

$$F_0 = f_{SFEB}(I_{LR}) \quad (1)$$

式中, $f_{SFEB}(\cdot)$ 是浅层特征提取函数, F_0 是其输出. 然后, F_0 再通过 FMB 进行深层特征提取:

$$F_n = f_{PFFG}^n(F_{n-1}), n = 1, 2, \dots, N \quad (2)$$

式中, $f_{PFFG}^n(\cdot)$ 是 FMB 中第 n 个 PFFG 函数, F_n 是其输出, F_{n-1} 是其输入即第 $n-1$ 个 PFFG 的输出, N 是 FMB 中 PFFG 的个数. 接着, 深层残差特征 F_N 组合浅层特征 F_0 , 通过上采样块重建目标 SR 图像:

$$I_{SR} = f_{UP}(F_n + F_0) \quad (3)$$

式中, $f_{UP}(\cdot)$ 是上采样块函数. 最后, 输出 SR 图像 I_{SR} .

2.1 渐进特征融合组

FMB 包含多个 PFFG, PFFG 中特征融合设计主要受到 Liu 等^[35] 启发. 他们的研究表明, 网络中不同深度的残差特征逐渐集中于输入图像的不同方面, 对于重建空间细节非常有用. 基于 CNN 的 SISR 模型大都只是将残差学习作为缓解训练难度的策略. SISR 模型堆叠残差块, 残差特征传播到下一个块之前, 与恒等特征融合, 致使后面的残差块只喂入了复杂的融合特征, 忽略了充分利用更清洁

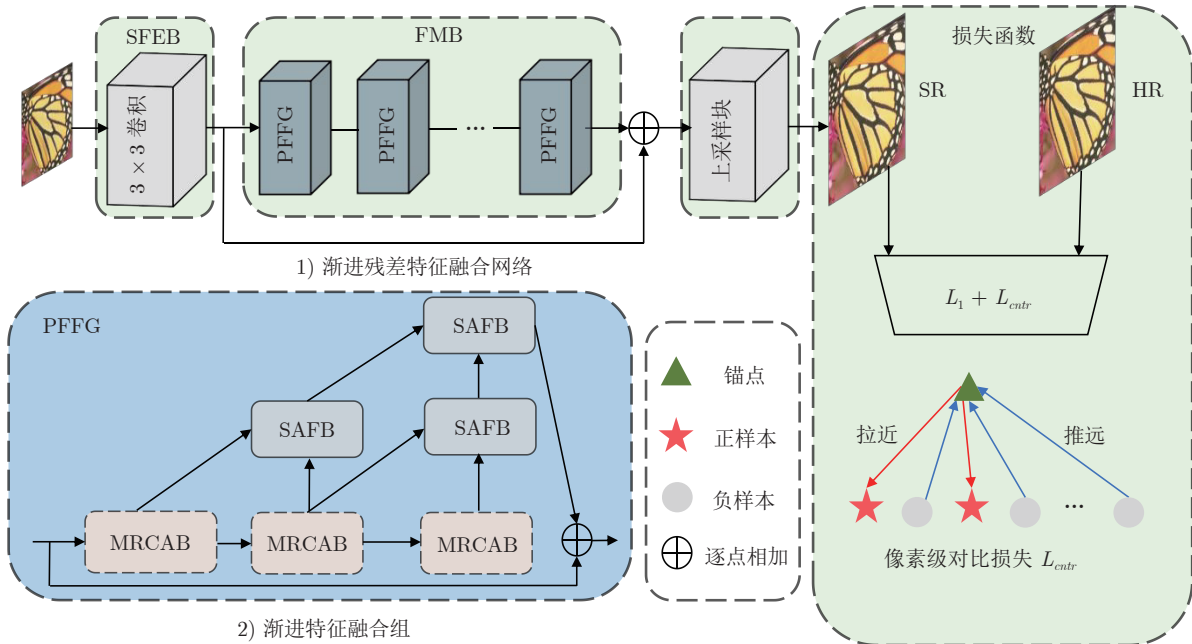


图 3 网络架构细节

Fig.3 Network architecture details

的残差特征. Liu 等^[35]设计的残差模块包含多个残差块, 并把各个残差块输出的特征拼接在一起, 馈入 1×1 卷积层进行融合.

PFFG 由 MRCAB 和 SAFB 组成, 每个 MRCAB 的输出特征是下一个 MRCAB 的输入, 且与后续所有 MRCAB 输出特征, 通过 SAFB 进行分层融合, 以强化中间残差特征, 见图 3 中模块 2). 每个 PFFG 使用了 3 个 MRCAB, 残差分支的特征融合可以分为 3 个步骤: 1) 从左到右, 每两个相邻的 MRCAB 输出特征通过 SAFB 进行融合; 2) 步骤 1) 输出的融合特征从左到右使用 SAFB, 对每两个相邻的特征进行融合, 其余类推; 3) 步骤 2) 输出的融合特征作为 PFFG 的残差分支输出, 与 PFFG 的输入特征求和. 每个 PFFG 有 3 个 MRCAB, 第 n 个 PFFG 中第 m ($m = 1, 2, 3$) 个 MRCAB 的输出为 $F_{n,m}^{MR}$, 则第 n 个 PFFG 中第 1 个 MRCAB 的输入是前一个 PFFG 的输出 F_{n-1} (第 1 个 PFFG 中, 第 1 个 MRCAB 的输入为 SFEB 的输出 F_0); 第 n 个 PFFG 中第 m ($m = 2, 3$) 个 MRCAB 的输入为前一个 MRCAB 的输出 $F_{n,m-1}^{MR}$. F_n 可表示为:

$$\begin{cases} F_{n,m}^{MR} = f_{MRCAB}(F_{n-1}), & m = 1 \\ F_{n,m}^{MR} = f_{MRCAB}(F_{n,m-1}^{MR}), & m = 2, 3 \end{cases} \quad (4)$$

式中, $f_{MRCAB}(\cdot)$ 是 MRCAB 函数; $F_{n,m}^{MR}$ 是第 n ($n = 1, 2, \dots, N$) 个 PFFG 中, 第 m ($m = 1, 2, 3$) 个 MRCAB 的输出:

$$F_n = f_{SAFB}(f_{SAFB}(F_{n,1}^{MR}, F_{n,2}^{MR}), f_{SAFB}(F_{n,2}^{MR}, F_{n,3}^{MR})) + F_{n-1} \quad (5)$$

式中, $f_{SAFB}(\cdot)$ 是 SAFB 块函数.

2.2 多尺度残差通道注意力块

本文的 MRCAB 主要受 Zhang 等^[11]和 Li 等^[25]启发. Li 等^[25]开发了一个多尺度的残差块 (Multi-scale residual block, MSRB). MSRB 包含两个分支, 一个分支使用 3×3 卷积, 另一个分支使用 5×5 卷积. 两个分支输出的不同尺度特征拼接在一起, 再分别通过 3×3 和 5×5 卷积. 通过实验可以发现, 更多尺度可以更好地利用特征信息, 因此本文的 MRCAB 使用三个分支, 如图 4 所示. 为了减少参数量, 一个分支使用 3×3 卷积, 其余两个分支使用 3×3 扩张卷积, 扩张率分别为 2 和 4. 另外, 为了简化 MSRB, 本文把三个分支输出的不同尺度特征简单地求和, 进行融合. 在 MRCAB 的后部, 引入了 Zhang 等^[11]的通道注意力机制, 通过自适应地伸缩各个通道, 以建模各个通道之间的依赖性. 第 n 个

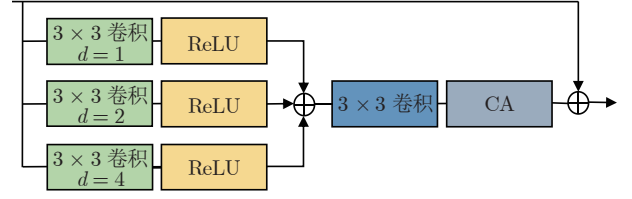


图 4 多尺度残差通道注意力块

Fig.4 Multi-scale residual channel attention block

PFFG 中, 第 m ($m = 1, 2, \dots, M$) 个 MRCAB 的第 k 个分支 ($k = 1, 2, 3$) 可表示为:

$$F_{S_m}^k = \sigma(DC_{3 \times 3}^d(F_{n,m-1}^{MR})) \quad (6)$$

式中, $F_{S_m}^k$ 是输出的特征, $\sigma(\cdot)$ 是 ReLU 非线性激活函数, $DC_{3 \times 3}^d(\cdot)$ 是 3×3 的扩张卷积 (扩张率 $d = 1, 2, 4$), $F_{n,0}^{MR} = F_{n-1}$.

第 n 个 PFFG 中, 第 m ($m = 1, 2, \dots, M$) 个 MRCAB 的输出可表示为:

$$F_{n,m}^{MR} = CA \left(\text{Conv}_{3 \times 3} \left(\sum_{k=1}^3 F_{S_m}^k \right) \right) + F_{n,m-1}^{MR} \quad (7)$$

式中, $CA(\cdot)$ 是 RCAN 中通道注意力函数.

2.3 空间注意力融合块

考虑到 MRCAB 中使用了 Zhang 等^[11]的通道注意力建模特征通道之间的依赖关系, 在融合两个 MRCAB 的输出特征时, 使用空间注意力, 进一步建模特征像素之间的依赖关系. 两个 MRCAB 的输出特征拼接在一起, 使用 1×1 卷积进行融合, 生成两个特征通道, 再使用 Sigmoid 函数, 将通道像素值转换到 (0, 1) 之间, 作为输入特征的权重. 学习输入特征像素之间的依赖关系, 如图 5 所示. 假定输入的特征分别为 F_A 和 F_B , SAFB 中特征融合过程可表示为:

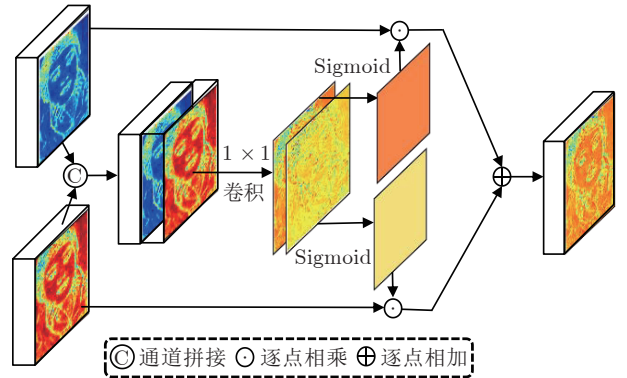


图 5 空间注意力融合块

Fig.5 Spatial attention fusion block

$$\begin{cases} W_A, W_B = f_{split}(\text{Conv}_{1 \times 1}([F_A, F_B])) \\ F_{fuse} = \text{Sigmoid}(W_A) \cdot F_A + \text{Sigmoid}(W_B) \cdot F_B \end{cases} \quad (8)$$

式中, F_{fuse} 为 SAFB 输出的融合特征, $\text{Conv}_{1 \times 1}(\cdot)$ 为 1×1 卷积函数, $f_{split}(\cdot)$ 把两个通道分裂为矩阵 W_A 和 W_B , $\text{Sigmoid}(\cdot)$ 为 Sigmoid 函数, “ $\cdot$ ” 为逐像素相乘。

SAFB 利用空间注意力机制, 渐进地融合分层的特征信息, 可以更好地利用和融合多样化的特征信息。

3 损失函数

本文的核心工作是引入像素级对比损失, 使被恢复的像素更靠近原 HR 图像中对应像素, 远离其他邻近像素, 以提高 SR 图像的保真度和清晰度. 对比学习框架^[29, 32, 36-37] 主要是为高层语义理解任务设计的, 在低级视觉中应用的潜力还没有得到充分探索. 对于给定的锚点, 对比学习的目标是将锚点拉向正样本, 远离负样本. 基于 CNN 的 SISR 方法虽然显著改进了之前的 SISR 方法, 但如何恢复 SR 图像中丢失、模糊或失真的细节, 仍然是有待解决的问题. 目前, 已有一些损失函数 (如感知损失 L_{perc} ^[15]、纹理损失 L_{tex} ^[19] 和对抗损失 L_{adv} ^[20] 等) 改进了被广泛使用的 L_1 或 L_2 损失, 但仍不能令人满意. L_{perc} 和 L_{tex} 仍然不能阻止模糊, L_{adv} 虽然能生成清晰的细节, 但生成的细节严重失真和不自然. 为了解决这个问题, 本文提出像素级对比损失 L_{cntr} . L_{cntr} 可以与 L_1 、 L_2 、 L_{perc} 、 L_{tex} 和 L_{adv} 等一个或多个结合使用, 进一步改进 SR 图像的保真度和清晰度。

优化 L_1 或 L_2 损失函数是将待恢复的 SR 图像中, 每个像素逐渐逼近原 HR 图像中对应的像素. 遗憾的是, SR 图像恢复是一个病态逆问题, 同一个低分辨率图像可以对应多个高分辨率图像, 恢复的 SR 图像是许多高分辨率图像的平均, 导致过度平滑. 本文的像素级对比损失 L_{cntr} 将 SR 图像中的像素作为锚点, HR 图像中对应的像素作为正样本, 与该正样本像素值相同的像素也作为正样本, 其他像素作为负样本. L_{cntr} 与 L_1 或 L_2 损失函数联合使用, 增加一个附加的约束, 让锚点接近正样本, 远离负样本, 具体过程见图 6. 不同的负样本可以自适应地推远锚点. L_{cntr} 的形式类似于 InfoNCE^[38] 损失, 它是基于 Softmax 的分类损失, 在小批量样本中, 分类正样本和负样本. 原 HR 图像分为许多大小相等的区域 (假定共有 Q 个区域), 在每个局部区域计算对比损失. 在给定 HR 图像中, 第 q 个局部区域假定包含 S 个像素, 第 i 个像素 (记为 $x_{HR,i}^q$) 视为

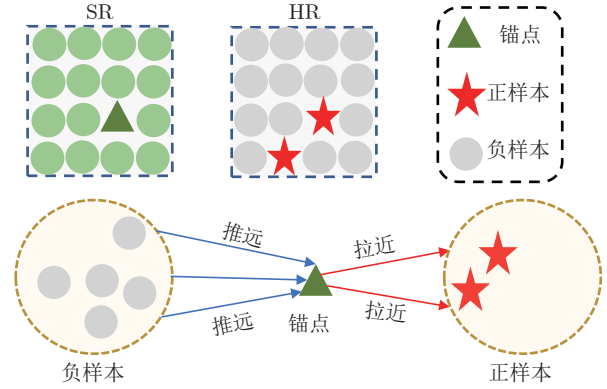


图 6 像素级对比损失

Fig.6 Pixel-wise contrastive loss

正样本, 其他具有相同值的像素, 也视为正样本, 该区域对比损失计算为:

$$L_{cntr}^q = -\frac{1}{S} \sum_{i=1}^S \ln \frac{\sum_{j=1}^{S_1} \exp(-\|x_{SR,i}^q - x_{P,j}^q\|_1)}{\sum_{j=1}^S \exp(-\|x_{SR,i}^q - x_{HR,j}^q\|_1)} \quad (9)$$

式中, L_{cntr}^q 为第 q 个区域对比损失, 即该区域中每个像素视为正样本时的平均值. $x_{SR,i}^q$ 为 SR 图像第 q 个区域中, 第 i 个锚点; HR 图像第 q 个区域中, 与之对应的 $x_{HR,i}^q$ 是正样本. 在该区域中, 与 $x_{HR,i}^q$ 值相同的像素也视为正样本, 记为 $x_{P,j}^q$, 总共 S_1 个 ($j = 1, 2, \dots, S_1$). 总逐像素对比损失 (表示为 L_{cntr}) 是 Q 个区域中, 逐像素对比损失 L_{cntr}^q 之和, 即:

$$L_{cntr} = \sum_{q=1}^Q L_{cntr}^q \quad (10)$$

网络模型总损失计算为:

$$L_C = \lambda_C L_{cntr} + L_1 \quad (11)$$

式中, λ_C 是系数, 以平衡 L_{cntr} 和 L_1 损失.

4 实验分析

4.1 实验设置

本文使用 DIV2K 数据集^[39] 训练模型, 其中 800 个图像作为训练图像, 5 个图像用于验证集, 标记为 DIV2K_val5. 为了增强训练图像, 训练图像进行了 90° 旋转和随机的水平翻转. 测试中, 使用 Set5^[40]、Set14^[6]、B100^[41]、Urban100^[42] 和 Manga-109^[43] 5 个标准测试数据集; 对原 HR 图像进行双三次下采样, 以获得对应的 LR 图像. 与其他方法类似, 使用 PSNR 和 SSIM 评估模型的客观性能, PSNR 和 SSIM 均在 YCbCr 空间^[23] 的 Y 通道上进行计

算. 训练中, 批大小设置为 16, LR 图像片尺寸设置为 48×48 像素. 使用 ADAM 优化器, 设置 $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$. 初始学习率设置为 10^{-4} , 每 200 个迭代周期减半. 使用 PyTorch 框架实现模型, 并在 NVIDIA GeForce RTX 2080Ti GPU 上训练.

4.2 实验分析

4.2.1 消融实验

为了验证本文网络架构和像素级对比损失的有效性, 进行了 5 组对比实验: 1) 把网络架构中的 MRCAB 替换为 RCAB, 仅使用 L_1 损失, 不包含 SAFB 块和像素级对比损失 L_{cntr} , 以该模型作为基准, 称为 PRFFN0; 2) 在 PRFFN0 基础上, 增加类似于残差特征聚合 (Residual feature aggregation, RFA)^[35] 的特征融合方法, 与 RFA 块的区别是将其残差子块替换为 RCAB, 这个模型称为 PRFFN1; 3) 将 PRFFN1 中的 RFA 特征融合方法替换为 SAFB, 进行特征融合, 这个模型称为 PRFFN2; 4) 将 PRFFN2 中的 RCAB 替换为 MRCAB, 这个模型称为 PRFFN3; 5) 在 PRFFN3 中, 增加像素级对比损失 L_{cntr} , 这个模型称为 PRFFN 即本文模型. 训练 1000 个迭代周期, 5 个模型在 DIV2K_val5 验证集上 3 倍超分辨率的平均 PSNR 和参数量如表 1 所示. 表 1 中, “✓”表示在训练时使用该损失, “—”表示训练时不使用该损失. 由表 1 可以看出, 基准模型 PRFFN0 的平均 PSNR 为 32.259 dB; RFA 特征融合改进了 PRFFN0 的 PRFFN1, 平均 PSNR 增加了 0.048 dB. 相较于 RFA, 本文 SAFB 使用了空间注意力机制, 增强了特征空间依赖关系之间的建模, 优于 RFA 中特征图的简单拼接, 使 PRFFN2 较 PRFFN1 模型, 参数量减少了 55 K, 平均 PSNR 却增加了 0.035 dB; 相较于 RCAB, 本文 MRCAB 使用了扩张卷积, 在不增加模型参数量的情况下, 扩大了特征感受野, 且融合了多个尺度扩张卷积特征, 使 PRFFN3 较 PRFFN2, 平均 PSNR 增加了 0.022 dB; 在 L_1 损失的基础上, 增加像素级对比损失 L_{cntr} , 提高了 SR 图像的保真度, 使 PRFFN 较 PRFFN3, 平均 PSNR 增加了 0.087 dB.

4.2.2 对比损失的设计与分析

1) 对比损失的有效性. 为了验证对比损失的有效性, 实验比较了 L_1 、 L_{perc} 、 L_{CSD} 和 L_{cntr} 损失及其组合对应的 PSNR 和学习的感知图像块相似性 (Learned perceptual image patch similarity, LPIPS)^[44] 度量. LPIPS 被认为更符合人类的视觉感知, 视觉感知质量随 LPIPS 的降低而增加, 而

表 1 DIV2K_val5 验证集上, 不同模型, 3 倍 SR 的平均 PSNR 和参数量

Table 1 The average PSNRs and parameter counts of 3 times SR for different models on the DIV2K_val5 validation data set

模型	L_1	L_{cntr}	参数量 (K)	PSNR (dB)
PRFFN0	✓	—	2975	32.259
PRFFN1	✓	—	3222	32.307
PRFFN2	✓	—	3167	32.342
PRFFN3	✓	—	3167	32.364
PRFFN	✓	✓	3167	32.451

PSNR 值越高, 则图像的保真度越高. 训练 1000 个迭代周期, 实验结果如表 2 所示, 表 2 中“✓”表示在训练时, 使用该损失; “—”表示训练时, 不使用该损失. L_{cntr} 与其他损失组合时, 可改进 PSNR 和 LPIPS 值. 当 L_{cntr} 与 L_1 组合时, 获得了最好的 PSNR 值 (32.451 dB); 当 L_{cntr} 与 L_1 、 L_{CSD} 组合时, 获得了最好的 LPIPS 值 (0.0613). 由表 2 可知, L_{CSD} 作用于图像特征空间, 获得了更好的 LPIPS; L_{cntr} 作用于像素空间, 获得了更好的 PSNR, 即保真度更高. L_{cntr} 组合 L_{CSD} 虽获得了最好的 LPIPS, 但 PSNR 有所降低.

表 2 DIV2K_val5 验证集上, 不同损失函数及其组合, 3 倍 SR 的平均 PSNR 和 LPIPS 结果

Table 2 The average PSNRs and LPIPSs of 3 times SR for different losses and their combinations on the DIV2K_val5 validation data set

L_1	L_{perc}	L_{CSD}	L_{cntr}	PSNR (dB)	LPIPS
✓	—	—	—	32.364	0.0978
✓	—	—	✓	32.451	0.0969
✓	✓	—	—	32.236	0.0672
✓	✓	—	✓	32.305	0.0656
✓	—	✓	—	32.387	<u>0.0624</u>
✓	—	✓	✓	<u>32.432</u>	0.0613

2) 不同比例对比损失影响. 如式 (11) 所示, 比例系数 λ_C 可以调节对比损失的比例. 训练 1000 个迭代周期, 在 DIV2K_val5 验证集上, 不同 λ_C , 3 倍 SR 的平均 PSNR 结果见表 3. 由表 3 可以看出, 过大或过小的 λ_C 使 PSNR 结果恶化. 当 $\lambda_C = 10^{-1}$ 时, 获得了最好的 PSNR 性能.

3) 对比损失的通用性. 本文的对比损失 L_{cntr} 可以与 L_1 、 L_{perc} 和 L_{CSD} 等损失组合使用, 从而改进 SR 图像的保真度. L_{cntr} 也可用于其他模型, 可以改进 SR 的性能. 以著名的 EDSR、RCAN 和 SwinIR-light^[26] 为例, 训练 1000 个迭代周期, 原模型和增加 L_{cntr} 损失后的平均 PSNR 和 SSIM 结果

表 3 DIV2K_val5 验证集上, 不同 λ_C , 3 倍 SR 的平均 PSNR 结果

Table 3 The average PSNRs of 3 times SR for different λ_C on the DIV2K_val5 validation data set

λ_C	PSNR (dB)
10^{-2}	32.386
10^{-1}	32.451
1	32.381
10	32.372

如表 4 所示, 表 4 中“ $\checkmark$ ”表示在训练时使用该损失, “—”表示训练时不使用该损失, “ $\uparrow$ ”代表指标提升. 由表 4 可以看出, L_{cntr} 损失可以显著改进 SR 图像视觉效果. SwinIR-light 增加 L_{cntr} 后, 在 Urban-100 数据集中, img004 图像的 3 倍 SR 结果如图 7 所示. 由图 7 可以看出, 使用 L_{cntr} 损失后, SR 图像的视觉质量得到显著改进.

表 4 DIV2K_val5 验证集上, 不同模型包含与不包含 L_{cntr} 损失, 3 倍 SR 的平均 PSNR 和 SSIM 结果

Table 4 The average PSNRs and SSIMs of 3 times SR for different models with and without L_{cntr} loss on the DIV2K_val5 validation data set

模型	L_{cntr}	PSNR (dB)	SSIM
EDSR	—	32.273	0.9057
	$\checkmark$	32.334 ($\uparrow 0.061$)	0.9067 ($\uparrow 0.0010$)
SwinIR-light	—	32.442	0.9062
	$\checkmark$	32.489 ($\uparrow 0.047$)	0.9069 ($\uparrow 0.0007$)
RCAN	—	32.564	0.9088
	$\checkmark$	32.628 ($\uparrow 0.064$)	0.9096 ($\uparrow 0.0008$)

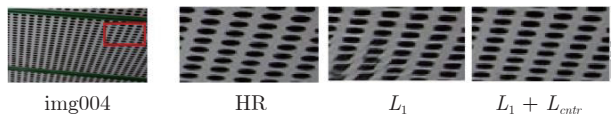


图 7 Urban100 数据集中, SwinIR-light 使用不同损失函数, img004 图像的 3 倍 SR 结果

Fig.7 The 3 times SR results of SwinIR-light using different losses on the img004 image in the Urban100 data set

4) 不同大小局部区域对对比损失的影响. 计算对比损失的方形局部区域, 可用区域中像素个数 S 度量. 本文考虑 4×4 像素、 8×8 像素和 16×16 像素的局部区域, S 分别为 16、64 和 256. 在 DIV2K_val5 验证集上, 训练 1000 个迭代周期, 不同 S 值的 3 倍 SR 的平均 PSNR 结果见表 5. 可以看出, 更大的局部区域 (即更大的 S 值) 负样本更多, 有利于改进 PSNR 性能. 但在训练时, 更大的 S 值需要更

表 5 DIV2K_val5 验证集上, 不同大小局部区域, 3 倍 SR 的平均 PSNR 结果

Table 5 The average PSNRs of 3 times SR for different size local regions on the DIV2K_val5 validation data set

S	PSNR (dB)
16	32.425
64	32.451
256	32.458

表 6 3 倍 SR 训练 10 个迭代周期, 训练占用的内存和使用的训练时间

Table 6 For 3 times SR, 10 epochs, comparing the memory and time used by training

L_{cntr}	S	内存 (MB)	时间 (s)
—	—	4126	1962
$\checkmark$	16	4836	2081
$\checkmark$	64	5216	2219
$\checkmark$	256	7541	2893

多的时间和更大的内存量 (见表 6). 考虑到相较于 $S = 64$, 在 $S = 256$ 时, PSNR 改进很小, 因此本文选择 $S = 64$.

5) 对比学习训练的计算代价. 4 组对比实验分别为不使用 L_{cntr} 损失以及 $S = 16$ 、64 和 256 三种情形. 对于 3 倍 SR 训练 10 个迭代周期, 训练占用的内存和使用的训练时间如表 6 所示. 表 6 中, “ $\checkmark$ ”表示在训练时使用该损失, “—”表示训练时不使用该损失. 由表 6 可以看出, 计算代价没有显著增加. 当 $S = 64$ 时, 内存增加了约 21%, 计算时间仅增加了约 11%.

4.2.3 PFFG 实验

在网络架构中, 共由 10 个 PFFG 组成, 每个 PFFG 包含 3 个 MRCAB 和 3 个 SAFB, 见图 3 中模块 2).

1) 扩张卷积的不同扩张率对性能的影响. 与 RCAN 中的 RCAB 相比, MRCAB 没有增加参数, 并且通过多分支提取了多尺度特征信息, 并通过扩张卷积 (或空洞卷积) 扩大了感受野. MRCAB 中, 3 个分支扩张卷积的扩张率组合分别设为 1, 2, 3、1, 2, 4 和 1, 2, 5. 考虑只有 1 个分支使用常规卷积即扩张率为 1, 此时 MRCAB 退化为 RCAB; 考虑 2 个分支, 扩张率组合为 1, 2 的情形, 训练 1000 个迭代周期, 不同分支和不同扩张率组合, 3 倍 SR 的平均 PSNR 结果见表 7. 由表 7 可见, 不同尺度的特征融合可以改进性能. 扩张率更大, 特征感受野大, 对性能有益. 但过大的扩张率, 也是有害的. MRCAB 三个分支的扩张率设置为 1, 2, 4 较为合理.

表 7 DIV2K_val5 验证集上, MRCAB 不同分支和不同扩张率组合, 3 倍 SR 的平均 PSNR 结果

Table 7 The average PSNRs of 3 times SR for the different branches of MRCAB with different dilation rate combinations on the DIV2K_val5 validation data set

不同的扩张率卷积组合	PSNR (dB)
1	32.375
1, 2	32.370
1, 2, 3	32.392
1, 2, 4	32.451
1, 2, 5	32.415

4.3 与先进方法比较

1) 客观比较. 本文与代表性的双三次上采样、快速超分辨率卷积神经网络 (Fast super-resolution convolutional neural network, FSRCNN)^[45]、稀疏掩码超分辨率网络 (Sparse mask super-reolu-

tion, SMSR)^[46]、自适应级联注意力网络 (Adaptive cascading attention network, ACAN)^[47]、权重自适应超分辨率网络 (Adaptive weighted super-resolution network, AWSRN)^[48]、深度递归卷积网络 (Deeply-recursive convolutional network, DRCN)^[49]、级联残差网络 (Cascading residual network, CARN)^[50]、OISR-RK2 (Apply ODE-inspired schemes to super-resolution network designs-Runge-Kutta)^[51]、OISR-LF (Apply ODE-inspired schemes to super-resolution network designs-leapfrog)^[51]、MSRN、软边缘辅助网络 (Soft-edge assisted network, SeaNet)^[52]和两阶段注意力网络 (Two-stage attentive network, TSAN)^[53] 方法进行比较. 在 5 个标准测试数据集上, 2 倍、3 倍和 4 倍 SR 的平均 PSNR 和 SSIM 结果见表 8. 与最先进的 TSAN 相比, 本文模型参数和计算量大约减少了 1/3, 推理时间大约减少了一半, 但性能更好. 例如, 对于 2 倍 SR, 本文方法与 TSAN 相当; 对于 3 倍和 4 倍 SR, 在所有测试数据

表 8 5 个标准测试数据集上, 不同 SISR 方法的 2 倍、3 倍和 4 倍 SR 的平均 PSNR 和 SSIM 结果

Table 8 The average PSNRs and SSIMs of 2 times, 3 times, and 4 times SR for different SISR methods on five standard test data sets

放大倍数	方法	参数量 (K)	计算量 (G)	推理时间 (ms)	Set5 PSNR/SSIM	Set14 PSNR/SSIM	B100 PSNR/SSIM	Urban100 PSNR/SSIM	Manga109 PSNR/SSIM
2	FSRCNN	13	6.0	26	37.00/0.9558	32.63/0.9088	31.53/0.8920	29.88/0.9020	36.67/0.9694
	SMSR	985	224.1	536	38.00/0.9601	33.64/0.9179	32.17/0.8990	32.19/0.9284	38.76/0.9771
	ACAN	800	2108.0	—	38.10/0.9608	33.60/0.9177	32.21/0.9001	32.29/0.9297	38.81/0.9773
	AWSRN	1397	320.5	506	38.11/0.9608	33.78/0.9189	32.26/0.9006	32.49/0.9316	<u>38.87/0.9776</u>
	DRCN	1774	17974.0	—	37.63/0.9588	33.04/0.9118	31.85/0.8942	30.75/0.9133	37.55/0.9732
	CARN	1592	222.8	352	37.76/0.9590	33.52/0.9166	32.09/0.8978	31.92/0.9256	38.36/0.9765
	OISR-RK2	4971	1145.7	715	38.12/0.9609	33.80/0.9193	32.26/0.9007	32.48/0.9317	38.79/0.9773
	OISR-LF	4971	1145.7	722	38.12/0.9609	33.78/0.9196	32.26/0.9007	32.52/0.9320	38.80/0.9774
	MSRN	6078	1356.8	810	38.08/0.9607	33.70/0.9186	32.23/0.9002	32.29/0.9303	38.69/0.9772
	SeaNet	7471	3709.1	1920	38.08/0.9609	33.75/0.9190	32.27/0.9008	32.50/0.9318	38.76/0.9774
3	TSAN	3989	1013.1	1183	38.22/0.9619	<u>33.84/0.9218</u>	32.32/0.9015	32.77/0.9345	—/—
	PRFFN	2988	656.4	792	<u>38.18/0.9611</u>	33.90/0.9207	<u>32.30/0.9012</u>	<u>32.75/0.9337</u>	39.02/0.9777
	FSRCNN	13	4.6	14	33.16/0.9140	29.43/0.8242	28.53/0.7910	26.43/0.8080	30.98/0.9212
	SMSR	993	100.5	281	34.40/0.9270	30.33/0.8412	29.10/0.8050	28.25/0.8536	33.68/0.9445
	ACAN	1115	1051.7	—	34.46/0.9277	30.39/0.8435	29.11/0.8065	28.28/0.8550	33.61/0.9447
	AWSRN	1476	150.6	263	34.52/0.9281	30.38/0.8426	29.16/0.8069	28.42/0.8580	33.85/0.9463
	DRCN	1774	17974.0	—	33.85/0.9215	29.89/0.8304	28.81/0.7954	27.16/0.8311	32.31/0.9328
	CARN	1592	118.8	177	34.29/0.9255	30.29/0.8407	29.06/0.8034	27.38/0.8493	33.50/0.9440
	OISR-RK2	5640	578.6	366	34.55/0.9282	30.46/0.8443	29.18/0.8075	28.50/0.8597	<u>33.80/0.9442</u>
	OISR-LF	5640	578.6	367	34.56/0.9284	30.46/0.8450	29.20/0.8077	28.56/0.8606	33.78/0.9441
4	MSRN	6078	621.2	476	34.38/0.9262	30.34/0.8395	29.08/0.8041	28.08/0.8554	33.44/0.9427
	SeaNet	7397	3233.2	994	34.55/0.9282	30.42/0.8444	29.17/0.8071	28.50/0.8594	<u>33.73/0.9463</u>
	TSAN	4174	565.6	650	<u>34.64/0.9282</u>	<u>30.52/0.8454</u>	<u>29.20/0.8080</u>	<u>28.55/0.8602</u>	—/—
	PRFFN	3167	312.5	441	34.67/0.9288	30.54/0.8460	29.23/0.8084	28.65/0.8621	34.03/0.9473

表 8 5 个标准测试数据集上, 不同 SISR 方法的 2 倍、3 倍和 4 倍 SR 的平均 PSNR 和 SSIM 结果 (续表)
Table 8 The average PSNRs and SSIMs of 2 times, 3 times, and 4 times SR for different SISR methods on five standard test data sets (continued table)

放大倍数	方法	参数量 (K)	计算量 (G)	推理时间 (ms)	Set5	Set14	B100	Urban100	Manga109
					PSNR/SSIM	PSNR/SSIM	PSNR/SSIM	PSNR/SSIM	PSNR/SSIM
4	FSRCNN	13	4.6	8	30.71/0.8657	27.59/0.7535	26.98/0.7150	24.62/0.7280	27.90/0.8517
	SMSR	1006	57.2	192	32.12/0.8932	28.55/0.7808	27.55/0.7351	26.11/0.7868	30.54/0.9085
	ACAN	1556	616.5	—	32.24/0.8955	28.62/0.7824	27.59/0.7379	26.31/0.7922	30.53/0.9086
	AWSRN	1587	91.1	188	32.27/0.8960	28.69/0.7843	27.64/0.7385	26.29/0.7930	30.72/0.9109
	DRCN	1774	17974.0	—	31.56/0.8810	28.15/0.7627	27.23/0.7150	25.14/0.7510	28.98/0.8816
	CARN	1592	90.9	121	32.13/0.8937	28.60/0.7806	27.58/0.7349	26.07/0.7837	30.47/0.9084
	OISR-RK2	5500	412.2	241	32.32/0.8965	28.72/0.7843	27.66/0.7390	26.37/0.7953	30.75/0.9082
	OISR-LF	5500	412.2	239	32.33/0.8968	28.73/0.7845	27.66/0.7389	26.38/0.7953	<u>30.76/0.9080</u>
	MSRN	6078	365.1	352	32.07/0.8903	28.60/0.7751	27.52/0.7273	26.04/0.7896	30.17/0.9034
	SeaNet	7397	3065.6	704	32.33/ <u>0.8981</u>	28.72/ <u>0.7855</u>	27.65/0.7388	26.32/0.7942	30.74/ <u>0.9129</u>
	TSAN	4137	415.1	452	<u>32.40/0.8975</u>	<u>28.73/0.7847</u>	<u>27.67/0.7398</u>	<u>26.39/0.7955</u>	—/—
	PRFFN	3131	200.1	316	32.43/0.8983	28.75/0.7857	27.70/0.7398	26.55/0.7974	30.93/0.9133

集上, 本文方法均优于 TSAN. 特别是在 Urban100 数据集上, 平均 PSNR 最大改进分别为 0.1 dB 和 0.16 dB.

2) 主观效果比较. 各种方法 SR 的视觉效果比较见图 8 ~ 10. 可以看出, 本文方法恢复的 SR 图像视觉质量优于其他方法. 例如, 对于 Urban100 数据

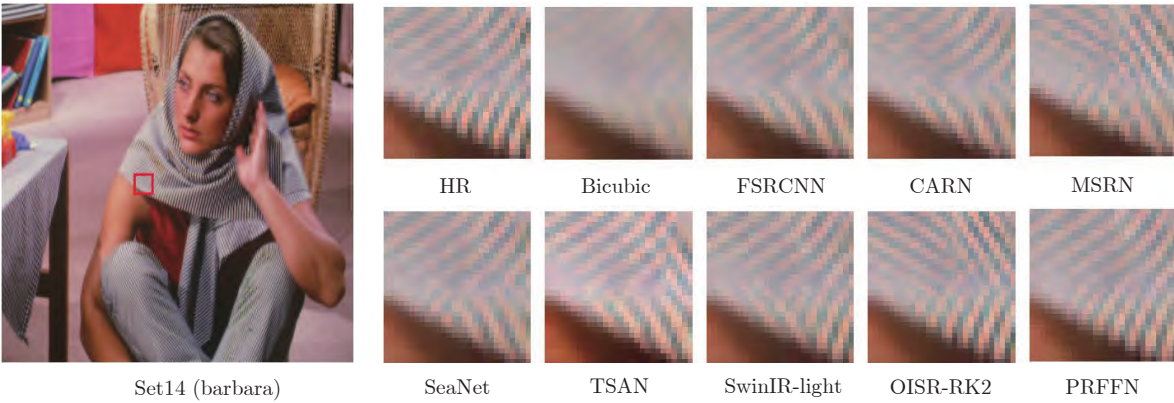


图 8 2 倍 SR 的视觉效果比较
Fig.8 Visual comparison for 2 times SR

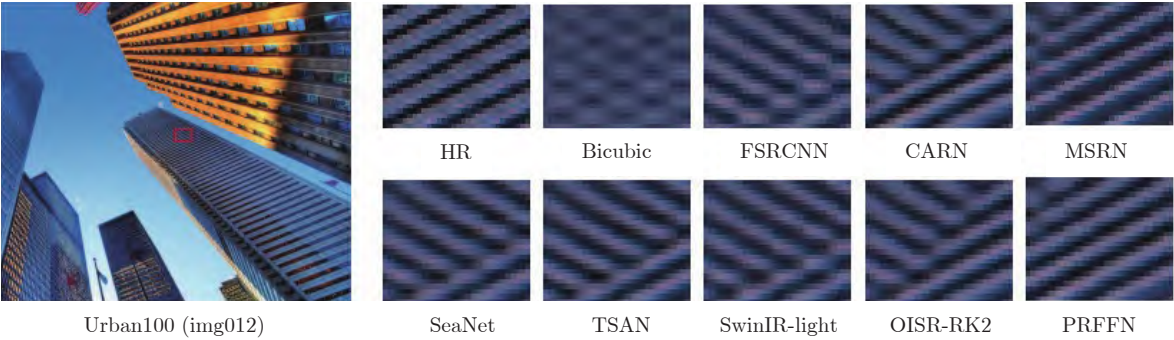


图 9 3 倍 SR 的视觉效果比较
Fig.9 Visual comparison for 3 times SR

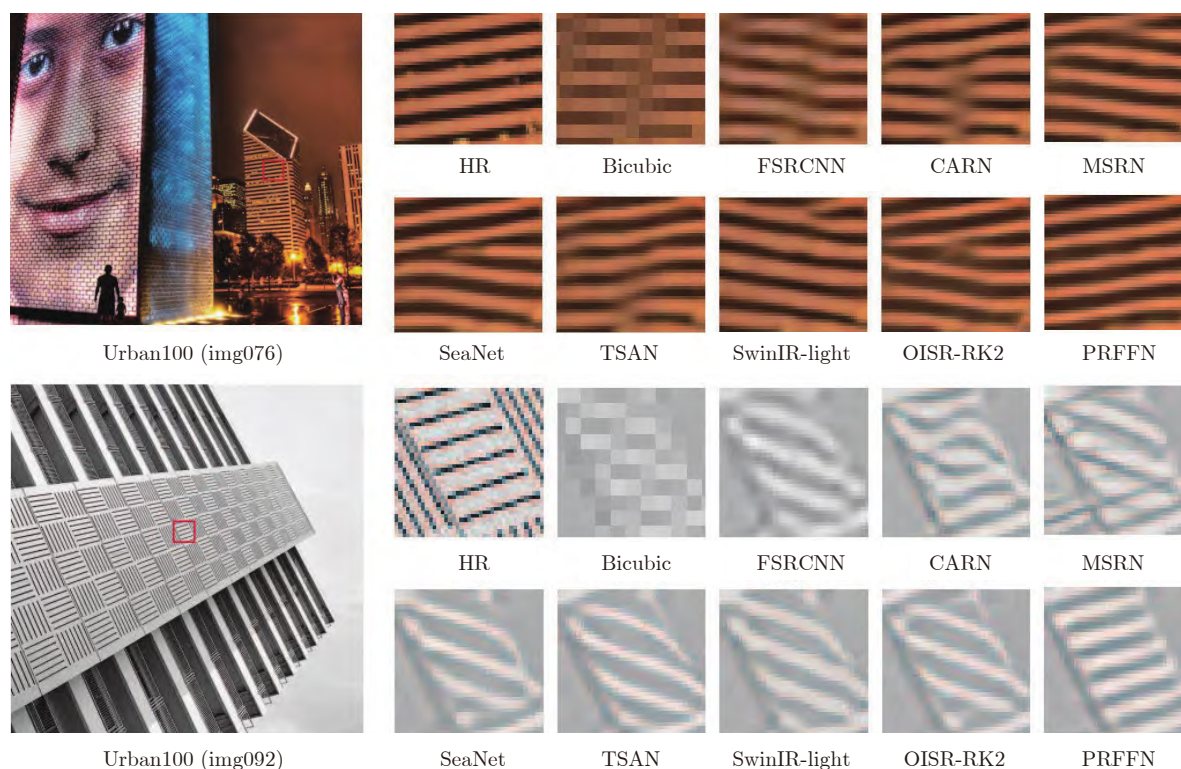


图 10 4 倍 SR 的视觉效果比较

Fig.10 Visual comparison for 4 times SR

集中的图像 img076, 其他方法恢复的 SR 图像存在过度模糊或条纹方向错误, 本文方法恢复出的 SR 图像接近于原 HR 图像; 其他图像的恢复结果也类似。

5 结束语

本文提出一个通用的基于对比学习的逐像素损失函数 L_{cntr} , 以恢复出的 SR 图像像素作为锚样本, 原 HR 图像对应的像素作为正样本, 其他像素作为负样本。 L_{cntr} 使锚样本尽可能接近正样本, 并远离负样本, 可以显著改进 SR 图像的保真度和视觉质量。为了验证 L_{cntr} 的性能, 本文提出一个 SR 的网络架构 PRFFN。实验结果表明, 本文的 PRFFN 组合 L_{cntr} 取得了很有竞争力的 SR 性能。

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